CYBERBULLYING DETECTION SYSTEM

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Certificate of Declaration

This is to certify that work entitled "**Cyberbullying Detection System**" *is submitted in partial fulfillment of the requirements for the award of Bachelor of Technology (Hons) in Software Engineering ,Harare Institute of Technology .It is further certified that no part of research has been submitted to any university for the award of any other degree .*

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HIT 400 /200 Project Documentation Marking Guide

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PRESENTATION- Format-Times Roman 12 for ordinary text, Main headings Times Roman 14, spacing 1.5.Chapters and sub-chapters, tables and diagrams should be numbered. Document should be in report form. Range of document pages. Between 50 and 100.Work should be clear and neat	**5**	
Pre-Chapter Section Abstract, Preface, Acknowledgements, Dedication & Declaration	**5**	
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Chapter Two-Literature Review Introduction, Related work & Conclusion	**10**	
Chapter Three –Analysis Information Gathering Tools, Description of system Data analysis –Using UML context diagrams ,DFD of existing system Evaluation of Alternatives Systems, Functional Analysis of Proposed System-Functional and Non-functional Requirements ,User Case Diagrams	**15**	
Chapter Four –Design Systems Diagrams –Using UML Context diagrams, DFD ,Activity diagrams Architectural Design-hardware, networking Database Design –ER diagrams ,Normalized Databases Program Design-Class diagrams, Sequence diagrams ,Package diagrams, Pseudo code Interface Design-Screenshots of user interface	**20**	
Chapter Five-Implementation & Testing Pseudo code of major modules /Sample of real code can be written here Software Testing-Unit, Module ,Integration ,System ,Database & Acceptance	**20**	
Chapter Six –Conclusions and Recommendations Results and summary ,Recommendations & Future Works	**10**	
Bibliography –Proper numbering should be used Appendices –templates of data collection tools, user manual of the working system, sample code, research papers	**5**	
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ABSTRACT

Cyberbullying has become a concerning issue in online game chat rooms. To tackle this problem, I have propose a Cyberbullying Detection System that uses Sentiment Analysis to identify cyberbullying behaviour in English language chat rooms. The primary objective of this system is to create a safe and secure online gaming environment by analyzing chat room messages for sentiment and producing activity reports.The system architecture has been meticulously designed to perform unsupervised and temporal sentiment analysis on text messages. This enables the early detection of cyberbullying instances and prompt implementation of preventative measures.

Moreover, the system has robust data security features such as key encryption and master signing mechanisms to safeguard user data. Overally, this system has the potential to significantly improve the safety and security of online game chat rooms by providing a strong defence against cyberbullying and ensuring the well-being of users in these virtual spaces.

PREFACE

Cyberbullying has become a widespread and concerning issue in recent years, with the rise of social media and online communication platforms. As more interactions take place online, the potential for harassment, abuse, and harmful behaviour has increased dramatically. This is a serious problem that can have devastating effects on the mental health and well-being of victims, especially young people.

The goal of this cyberbullying detection system project is to develop effective techniques and tools to identify and mitigate cyberbullying in online communities. Through the application of natural language processing, machine learning, and other advanced analytics, this system aims to automatically detect and flag potentially abusive or harassing content. The ultimate objective is to provide a means of creating safer, more inclusive online spaces where users can interact without fear of being targeted.

This preface outlines the motivation and high-level objectives of the cyberbullying detection system project. The subsequent chapters will delve into the technical details of the system's design, the datasets and methodologies employed, and the results of empirical testing and evaluation. It is our hope that this work will contribute meaningfully to the ongoing efforts to address the critical issue of cyberbullying.

ACKNOWLEDGEMENT

Firstly, I would like to thank **God** for taking me this far. I am grateful for the gift of life, good health and all the resources that I used to complete this project. Then, I specifically want to acknowledge **Mr Makondo**, my supervisor for the tremendous support, useful comments, remarks, feedback and engagement through the learning process of this capstone research project.

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I am indebted to my family for their unending support throughout this period. They gave me encouragement, love, guidance and support. May God bless them and protect them.

Lastly, I also extend my gratitude to my fellow classmates and friends who were there for me during this journey.

DEDICATION

This cyberbullying detection system is dedicated to all those who have been impacted by the harmful effects of cyberbullying. It is my sincere hope that this technology can help identify and mitigate online harassment, abuse, and bullying, so that everyone can feel safe and empowered to use the internet and digital platforms without fear.

Cyberbullying has devastating consequences, leading to anxiety, depression, low self-esteem, and in tragic cases, self-harm or suicide. No one should have to endure such cruelty. With this project, I aim to give victims of cyberbullying a voice and the tools to stand up to their tormentors.

It is my belief that through continued research, innovation, and a commitment to creating more compassionate online communities, we can turn the tide against digital abuse. This work is dedicated to a future where kindness, empathy, and mutual understanding prevail over harassment and hate.

Declaration

All sentences or passages quoted in this dissertation from other people's work have been specifically acknowledged by clear cross-referencing to author, work and page(s). I understand that failure to do this amount to plagiarism and will be considered grounds for failure in this dissertation and the degree examination as a whole.

CHAPTER 1 : Introduction

1.0 Introduction

Cyberbullying is an oppression happening virtually using devices such as computers, mobiles, and tablets. Cyberbullying can take place through messaging or on the internet in forums, social platforms, or gaming where community can share and post their thoughts. In short, social media are being used by bullies to harass people. Bullying can be analysed by consecutive behaviour and a purpose to harm which leads to having suicidal ideation, emotional responses and lower self-esteem such as the victim being angry, frustrated, scared and depressed. Children, in today's world, want their own mobiles and tablets at an adolescent age, and desire to connect to the social media platforms, and play online games such as Fortnite. If their behaviour goes unmonitored by their parents, then it may lead to cyber bullying. Examples of cyberbullying can include rumours posted on social media or sent by e-mail; embarrassing videos or pictures; and insulting, intimidating and abusive messages posted on social networks. When a message or a picture is posted online, it is very difficult to track and remove the content from the social media

The major problem with cyberbullying research is the lack of data, since researchers are traditionally forced to rely on survey data where victims and perpetrators self-report their impressions. In this paper, an automatic data collection system is presented that continuously collects in-game chat data from popular online games .The data was collected and combined with other information about the players from available online data services. The data collected is then analysed using analsyis and if the sentiments are negative the data is filtered removing data that contains negative sentiments. Classification of the collected data was carried out using simple feature detection with SQL database queries and compared to classification from AI-based sentiment text analysis services that have recently become available and further against manually classified data using a custom-built classification client built for this paper. Simple SQL classification and combined with modern machine learning algorithms will be used to identifying some features of toxic chat such as the use of bad language or racist sentiments.

1.2 Background

The surge of cyberbullying in online game chat rooms has become a major issue, requiring the development of effective strategies for detection and prevention. This thesis proposes a Cyberbullying Detection System that uses Sentiment Analysis to identify instances of cyberbullying in English language game chat rooms. The system's main objective is to establish a secure online gaming ecosystem by

scrutinizing sentiments expressed in game chat rooms. The system generates comprehensive reports to promote a culture of safety and responsible interaction among users. The system's architecture has two critical functions: unsupervised sentiment analysis and temporal analysis of text messages. These capabilities allow the system to identify and predict instances of cyberbullying, taking prompt preventative measures to mitigate harm and foster a respectful environment. To safeguard user data, the system has robust data security features such as key encryption and master signing mechanisms that protect user information from potential threats or breaches.

In summary, this visionary Cyberbullying Detection System has the potential to elevate safety and security standards within online game chat rooms, serving as a significant defence against cyberbullying. By doing so, it takes a step towards preserving the psychological and emotional well-being of users in virtual spaces, highlighting the importance of this pioneering endeavour in the digital landscape.

1.3 Problem Statement

This project aims to create a unique Cyberbullying Detection System that can identify instances of cyberbullying in the English language. This system has the potential to significantly impact the safety and well-being of users in online game chat rooms. The main purpose of this system is to act as a vigilant guardian, constantly monitoring and responding to cyberbullying incidents. With the help of advanced technological tools, such as sentiment analysis and real-time monitoring, it can provide a comprehensive shield against harmful online behaviours.

In the virtual world, anonymity and distance can often lead to fear and intimidation. This system aims to create a safe and secure environment for users to interact in, free from any such fears. It seeks to enrich their online gaming experience by fostering an environment where they can interact without any fear of cyberbullying.

1.4 Aims and Objectives

Aim: To create a system that will analyse sentiments in online games chat boxes to reduce cyber bullying

O**bjectives**: To realize this ambitious goal, the following precise objectives have been delineated:

- To perform sentiment analysis on text messages to predict cyberbullying in game chatrooms
- To notify the Admnistrator when cyberbullying text is detected.
- To block and delete cyberbullying text

- To generate a report of the activites in the system

These objectives represent a comprehensive strategy to build a system that promotes a secure and respectful online gaming environment. Additionally, this system provides users with effective tools to combat cyberbullying. This initiative reflects a strong commitment to the safety and well-being of individuals navigating the complex digital landscape.

1.5 Proposed Solution

The proposed system is aimed at detecting cyberbullying through sentiment analysis, making use of machine learning technology. The system will consist of three main modules: the machine learning model, the chat module (based on Discord), and the admin module for reports and database management.

1. The machine learning model is responsible for analyzing the natural language used in the Discord chat server and determining its polarity, or in other words, whether it is abusive or not.

2. The chat module is the platform where gamers can socialize and chat with each other. In this instance, the chat module will be based on Discord.

3. The Admin module will manage the database and display all messages that have been flagged as cyberbullying content.

The proposed system would operate in the following manner:

1. The Admin creates a game room server and a chatbot that automatically listens and monitors every activity in the server. The bot API is then assigned to our system.

2. The bot forwards every message to the system, which analyzes the polarity of the message and its intent. If both are negative, the system prompts the bot to delete the messages and record the message and user of the message that was flagged and deleted into the database.

3. The system generates reports that display all the flagged messages, their users, and timeframes.

1.6 Hypothesis

A Cyberbullying Detection System utilizing natural language processing (NLP) techniques and machine learning algorithms will effectively identify instances of cyberbullying by analyzing online text content. The system will outperform existing approaches in terms of accuracy, precision, and recall, thereby

providing an efficient and reliable solution for detecting and combating cyberbullying in various online platforms.

1.7 Justification

NLP Techniques: By employing advanced NLP techniques, such as sentiment analysis, semantic analysis, and contextual understanding, the Cyberbullying Detection System will be able to capture the nuances and subtleties of online text, enabling accurate identification of cyberbullying instances. This hypothesis assumes that these techniques will be capable of discerning the underlying negative intent and harmful content that characterizes cyberbullying.

Machine Learning Algorithms: The integration of machine learning algorithms, such as supervised learning models trained on labelled cyberbullying datasets, will enable the system to learn patterns and features indicative of cyberbullying behaviour. With sufficient training data and appropriate algorithm selection in this case I considered using the Naïve Bayes, the hypothesis assumes that the system will be able to generalize well and accurately classify instances of cyberbullying.

Comparative Performance: The hypothesis posits that the Cyberbullying Detection System will outperform existing approaches in terms of accuracy, precision, and recall. This assumption is based on the expectation that the system's combination of NLP techniques and machine learning algorithms will enable it to handle the complexities of cyberbullying detection more effectively, resulting in higher detection rates and lower false positive rates. Comparisons against existing systems or benchmark datasets would be necessary to validate this hypothesis.

Adaptability and Scalability: The hypothesis assumes that the Cyberbullying Detection System will be adaptable and scalable to different online platforms, including social media platforms, forums, chat rooms, and messaging apps. The system would need to demonstrate robustness in handling diverse textual data sources, user demographics, languages, and evolving cyberbullying techniques.

Ethical Considerations: The hypothesis acknowledges the importance of ethical considerations in the development of the Cyberbullying Detection System. It assumes that the system will be designed and trained in a manner that respects user privacy, avoids bias and discrimination, and minimizes false accusations while maximizing the detection of genuine cyberbullying instances.

Overall, the hypothesis proposes that a Cyberbullying Detection System leveraging NLP techniques and machine learning algorithms will provide an effective solution for accurately and efficiently identifying

instances of cyberbullying across various online platforms, outperforming existing approaches in terms of performance and adaptability.

1.8 Project Rationale

In the evolving landscape of social media platforms, including prominent platforms like Facebook and Twitter, the prevailing methods predominantly employ post-factum filtration, wherein content is scrutinized after it has already been posted – a paradigm often referred to as posterior actions. Concurrently, the absence of a comprehensive real-time system for the automatic detection of cyberbullying presents a critical gap, allowing many users to potentially encounter and interact with such posts before they are flagged and reported.

It is imperative to recognize that the scourge of cyberbullying transcends geographical boundaries and linguistic confines. This menace is a global phenomenon that spans diverse languages and cultures. Given its potential for profound and long-lasting impacts on victims, the imperative arises for an automatic detection system capable of identifying cyberbullying across various languages and locales. Such a system can play a pivotal role in proactively intervening and raising awareness by issuing warning messages to potential aggressors.

While it is true that existing approaches to automatic cyberbullying detection have primarily centered on English text and associated forums due to their prevalence, the sheer magnitude of users participating in online interactions necessitates a broader and more inclusive approach. The establishment of a comprehensive automatic cyberbullying detection system that transcends linguistic barriers is not merely an aspiration but a pressing need in today's digitally connected world.

As we delve deeper into the intricate landscape of cyberbullying detection methodologies, it becomes evident that a concerted effort is required to adapt and evolve these systems to encompass diverse languages and global contexts, thereby fortifying the defense against this pervasive online menace.

Scope of the project

The scope of this project serves as a comprehensive roadmap that delineates the precise contours of what must be achieved and the concerted efforts required to accomplish the overarching objectives. It encompasses an intricate tapestry of project goals, deliverables, features, functions, tasks, deadlines, and the associated costs, thus orchestrating a meticulous and systematic approach to project execution.

1. Project Goals: The foremost ambition of this project is to conceive, design, and implement a state-of-the-art Cyberbullying Detection System, characterized by its capacity to analyze sentiments within the confines of online gaming chat rooms. By achieving this primary goal, the project aspires to significantly diminish the prevalence of cyberbullying in virtual gaming spaces.

2. Deliverables: The project will yield several key deliverables, including but not limited to:

- A fully functional Cyberbullying Detection System.
- Comprehensive reports detailing user activities within the system.
- An unsupervised and temporal sentiment analysis module.
- Enhanced data security measures, featuring key encryption with master signing.

3. Features and Functions: The system's core features encompass:

- Sentiment analysis within online gaming chat rooms.
- Real-time reporting and monitoring of user interactions.
- Unsupervised and temporal sentiment analysis for cyberbullying prediction.
- Robust data security mechanisms to safeguard user information.

4. Tasks: The project encompasses a spectrum of tasks, ranging from system design and development to testing, deployment, and ongoing maintenance. These tasks will be meticulously executed to ensure the successful realization of project objectives.

5. Deadlines: The project timeline will be punctuated with well-defined deadlines and milestones, ensuring that each phase of development proceeds in a systematic and timely manner. This rigorous adherence to deadlines is imperative for project success.

6. Costs: An integral aspect of project scope, the financial dimension encompasses budgetary considerations, resource allocation, and cost estimates for development, deployment, and maintenance. A judicious financial plan will be instituted to manage project expenditures effectively.

In summation, the scope of this project not only serves as a guiding compass but also embodies a commitment to excellence and innovation in addressing the pressing issue of cyberbullying within the

realm of online gaming. It is a meticulously crafted blueprint that paves the way for the creation of a transformative system, ultimately ensuring a safer and more secure virtual space for all users.

1.9 Methodology

I proposed a model that is inspired by the observation that social media communication, like Twitter, often contains emotional signals, such as emoticons and specialized punctuation. These signals can be strongly associated with the sentiment of a social media post or its words. My aim is to leverage the emoticons used in game room chats and text messages to determine the overall sentiment score of the entire chat.

According to Sentiment Consistency Theory, there is a strong correlation between the sentiment polarity (positive, negative or neutral) of a post and the corresponding emotion signals. To model post-level emotion indication, I needed to ensure that the sentiment polarity of a post matches the emotion indication as closely as possible. The overall sentiment of a post is also positively correlated with the sentiments of the words used in that post. By modeling word-level emotional signals, I used the valuable information in the sentiment analysis framework to infer the sentiment polarity of a post.

To model post-level emoticon indication, I can build a classifier ($y = f(\mathrm{x})$), where y is a value between 0 and 1 indicating the sentiment indicated by the emoticons in the post, and x represents the list of features that can be extracted from game chat rooms.

The next step involves applying widely used classifiers such as Naïve Bayes to develop a sentiment prediction model. Logistic regression is the preferred empirical classifier as it is simple to train and understand, yet effective. To build the proposed model, posts with both emoticons and text are required to learn model parameters. This enables the model to predict the sentiment of posts that do not contain emoticons. The predicted sentiment score is represented by y^ which ranges from 0 to 1. The higher the predicted value, the more positive the sentiment.

I evaluated the effectiveness of the extracted features for sentiment prediction, and for this purpose, I used cluster analysis to assess the performance of the unsupervised sentiment model. As there is no access to ground truth, I could not compute standard accuracy measures. However, cluster analysis

enables to measure the quality of the patterns discovered by the unsupervised sentiment analysis. I was aiming to answer the following questions:

(1) Are the proposed sentiment features capable of clustering all the messages into distinct clusters that match the intuitive understanding of sentiment?

(2) What is the appropriate cluster size that should be used to decide the sentiment degree?

I used historical datasets from Kaggle to develop a logistic regression classifier for each event of cyberbullying. The features are the aggregate sentiment scores per tag/topic of interest, and the classes are the probability of a cyberbullying event occurring. The dataset has been divided into a training set and a testing set. The training set includes the first 80% of the time period the data covers, and the test set includes the other 20%. After developing the model, I implemented it on test game chatrooms to check its performance.

1.10 Feasibility Study

This examines the viability of an idea with a focus on identifying potential difficulties and attempts to answer questions such as: will the idea work? Should you go ahead with it? These will be examined to see if they are technically feasible given the resources available.

1.10.0 Technical Feasibility

The assessment for cyberbullying detection using sentiment analysis is an important step in the development of the system. The assessment seeks to answer questions such as whether there is supporting infrastructure and hardware already in place to support the development and implementation of the proposed system. It also aims to evaluate whether the developer will be able to develop a system that meets the user requirements.

The assessment may involve evaluating the existing computing resources and storage capacity to support the processing and storage of social media data for sentiment analysis. This could include examining any specialized hardware or software needed to perform the sentiment analysis, such as natural language processing (NLP) tools.

The assessment may also evaluate the developer's technical expertise and ability to build a system that meets the user requirements. This could include examining the developer's previous experience in developing similar systems, as well as their understanding of the user needs and how the system will be used.

Overall, the assessment plays a crucial role in determining the technical feasibility of the cyberbullying detection system and ensuring that it can be developed and implemented successfully.

1.10.1 Legal Feasibility

In Zimbabwe, the cyber law has recently been passed by the judicial system, which supports the automation and computerization of major processes. This makes the project for cyberbullying detection using sentiment analysis legally viable, as it aligns with the government's efforts to promote the use of technology in various fields.

With the passage of the cyber law, the legal framework is now in place to support the development and implementation of the cyberbullying detection system. This means that the project can proceed without any legal hurdles, as long as it adheres to the regulations and guidelines outlined in the cyber law.

Furthermore, the support for automation and computerization in Zimbabwe suggests that there is a growing acceptance and interest in the use of technology to improve various aspects of society. This bodes well for the success of the cyberbullying detection system, as it demonstrates a positive attitude towards technology adoption and innovation.

Overally, the legal viability of the project is an important factor to consider, and the recent passage of the cyber law in Zimbabwe provides a favorable environment for the development and implementation of the cyberbullying detection system using sentiment analysis.

1.10.2 Economic Feasibility

Using open source technologies mainly makes the project to be relatively cheap to implement. The table below shows the costs and expected benefits for economic feasibility.

Table 1 : Development Costs

Development Costs

Item	Cost ($)
Components	300
Web browser	Open Source
MySQL , Python ,Django Server	Open Source
Apache Web Server	Open Source
Hypertext Pre-processor	Open Source
Total Costs	$300

Installation and Operational Costs

Item	Cost ($)
Hardware maintenance	150
Software maintenance	200
Hosting Service	400
Accessories and other expenses	300
Total Costs	1050

Operational Feasibility

Cyberbullying is a growing problem that can have significant negative impacts on individuals, especially young people. Developing an effective system to detect and address cyberbullying is an important goal. This operational feasibility study examines the key considerations and requirements for implementing a cyberbullying detection system.

1.10.2 Operational Requirements

The core functional requirements of the cyberbullying detection system are:

1. **Content Monitoring**: The system must continuously monitor online content, including social media posts, messages, comments, etc. for potential cyberbullying behavior.
2. **Threat Assessment**: The system must analyze the monitored content and assess the severity and risk level of potential cyberbullying incidents.

3. **Reporting & Alerting**: The system must provide mechanisms for reporting detected cyberbullying cases to appropriate authorities or support services.
4. **Automated Moderation**: Where feasible, the system should have the capability to automatically flag and remove abusive content.
5. **User Support**: The system should provide resources and support for victims of cyberbullying, including access to counselling, legal aid.

1.11 PROJECT PLAN

A project plan is a formal document that provides guidance for the control and execution of a project. It is the most crucial document that needs to be created when starting any business project as it is the key to a successful project.

Table 2 : Project Plan

PHASE	**DURATION(weeks)**	**STARTING DATE**	**ENDING DATE**
Requirements analysis and definition	4 weeks	1 September 2022	30 September 2022
System design and software design	11 weeks	1 October 2022	05 Febrary 2023
Implementation and unit testing	3 weeks	06 January 2023	26 February 2023
Integration and system testing	4 weeks	27 February 2023	27 March 2023

1.12 GANTT CHART

Gantt charts are a visual tool that presents all the tasks required for a project and their sequence in relation to a timescale. This allows you to easily understand the project's structure, the tasks involved,

and their deadlines. It is an effective way to get a quick overview of a project and its progress.FigFFKJ

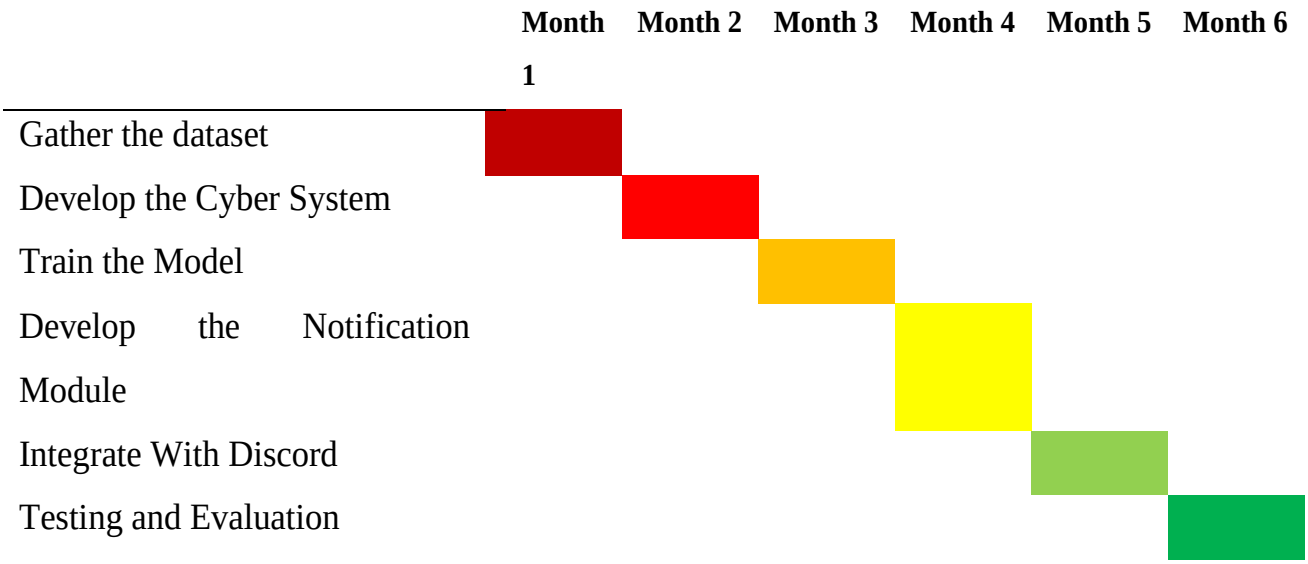

Fig 1 : Gantt Chart

CHAPTER 2 : Literature Review

2.0 Introduction

This chapter aims to discuss different sections that reflect relevant literature in relation to the current study. I will cover the theoretical framework, which includes various theories underlying the system to be developed. Additionally, I will look at the conceptual framework, which showcases different available concepts. Lastly, I will discuss the empirical framework, which lays out the current empirical evidence and findings based on the research under consideration. The purpose of the justification for this study is to make the significance of the research evident.

2.1 Related Work

There are various content filtering software systems available in the market that can be used by parents to monitor their children's social media activity and help detect and prevent cyberbullying. Some of the popular software systems include BullyBlocker, SafeChat, Facebook WatchDog, Rethink, Bark, and Bull Stop. These software systems are specifically designed to filter and analyze the online content to ensure the safety of children from online harassment and cyberbullying.

2.2 Bark

The Bark Application is a mobile app that was developed by Bark Technologies, Inc. for both Android and iOS devices (Bark Technologies, 2018). Parents can download the Bark Application from Google Play Store for Android devices or iTunes for iOS devices. The app is designed to enable parents to keep an eye on their children's social media activity. It can be connected to nearly 24 different social media websites, including popular ones such as Facebook, Twitter, and Instagram (T. Padmapriya et al). The application notifies parents if their children's social media accounts contain signs of cyberbullying, sexual implications, depression, and more.

2.3 BullyBlocker

BullyBlocker is an innovative solution to the problem of cyberbullying. It is an app that uses a sophisticated computational model to analyze social media platforms such as Facebook, to identify instances of cyberbullying. Once cyberbullying is detected, the app sends a notification to a parent or teacher so that appropriate action can be taken.

The app's computational model makes use of natural language processing and machine learning algorithms to analyze the text of social media posts and comments. It looks for patterns of language that are commonly associated with cyberbullying, such as insults, threats, and offensive language.

One of the key features of BullyBlocker is its ability to compute the intensity of cyberbullying. The app

uses a scale to measure the severity of each instance of cyberbullying, ranging from mild to severe. This information can be used to help parents and teachers understand the nature of the cyberbullying and take appropriate action. BullyBlocker is designed to be easy to use and customizable. Parents and teachers can set up the app to monitor specific social media accounts or keywords, and they can adjust the intensity threshold for cyberbullying detection.

Overall, BullyBlocker is a powerful tool for preventing cyberbullying and protecting young people from the harmful effects of online harassment. By using advanced computational techniques, the app provides a valuable service to parents, teachers, and anyone else concerned about the safety and well-being of children in the digital age.

2.4 Pocket Guardian

The Pocket Guardian Application is a mobile app that was created for iOS and Android by Pocket Guardian LLC (Llc 2018). For Android devices, the Google Play Store offers the application for download, and for iOS devices, iTunes offers the same. Parents can monitor their children's social media accounts and cell phones without violating their privacy by using the Pocket Guardian program. This app will only notify parents in the event that they find evidence of sexual or cyberbullying on their children's cell phone.

2.5 SafeChat

SafeChat is a software tool that helps protect children's communication on Facebook from explicit messages. The creators of SafeChat have developed a powerful model that uses natural language processing algorithms to identify specific words and phrases that are commonly associated with explicit content. The SafeChat model works by analyzing the text of Facebook messages and searching for keywords and phrases that may indicate the presence of explicit content. If the model detects any such content, it immediately blocks the message before it can be delivered to the recipient.

SafeChat is an invaluable tool for parents who are concerned about the safety of their children on social media. By preventing explicit messages from reaching children, SafeChat helps to protect them from harm and ensures that their online interactions are safe and appropriate. One of the key advantages of SafeChat is its ability to be customized to meet the needs of individual users. Parents can set up the tool to monitor specific Facebook accounts and adjust the sensitivity of the model to ensure that it accurately detects explicit content without generating false positives.

Overally, SafeChat is an effective solution to the problem of explicit content on social media. By using advanced natural language processing algorithms, the tool provides a powerful layer of protection for children and helps to ensure that they can communicate safely and securely on Facebook.

2.6 Bully Alert

The Bully Alert app is a mobile application developed by the CU Cyber Safety Lab as a research project for the University of Colorado Boulder (Lab 2018). This app is designed for Android and can be downloaded from the Google Play Store. It allows anyone to monitor another person's social media account and receive notifications if cyberbullying is detected on that account. Currently, the app only works with Instagram accounts. To use it, simply input the Instagram profile name of the account you want to monitor into the app. The app will then begin monitoring the profile and send a notification if any cyberbullying occurs on that account. The notification will provide details and context about the message being sent from that Instagram profile.

2.7 FacebookWatchDog

Facebook WatchDog is a software tool designed to detect threats and cyberbullying on social media. Its creators have developed a powerful system that uses social media analytics, image analysis, and text mining to identify potential threats and cyberbullying on platforms like Facebook.The tool analyzes social media activity by looking for patterns of behavior that may indicate the presence of a threat or instance of cyberbullying. This includes analyzing posts and comments for specific keywords and phrases that are commonly associated with threatening behavior or cyberbullying.

Facebook WatchDog uses image analysis to detect potential threats by analyzing images on social media platforms and looking for signs of violence, weapons, or other potentially dangerous content.The software tool also incorporates social media analytics to identify patterns of behavior that may be indicative of a potential threat or cyberbullying. This includes analyzing the frequency and intensity of social media activity, as well as the social networks and connections of individual users.

Overally, Facebook WatchDog is a powerful tool for protecting individuals from threats and cyberbullying on social media. Its advanced analytics and image analysis techniques provide a valuable layer of protection and help ensure that individuals can use social media platforms like Facebook safely and securely.

2.8 Bull

Stop Bull Stop application is a mobile application developed for Android devices as part of a Ph.D. study at the University of Aston, United Kingdom (Salawu, 2020). The application is designed to help young people combat cyber bullying proactively, with its target audience being children aged 13 above. The application works by using a deep learning-based algorithm to analyse messages and flag any offensive content such as cyber bullying, abuse or insults. For this application, if the application detects some

offensive contents in the user's social media account, it will automatically delete the messages to prevent them from reaching the user. Bull Stop application also allows users to block other user's social media from the application directly. This application currently only works on Twitter social media account with plans for Facebook and Instagram in the future

2.9 Rethink

This app can be installed on any device. As soon as a user types some word, this app runs in the background and looks for abusive word. It then shows a warning message, if that word belongs to bullying category.Although these software can provide a method for parents to keep an eye on children's activity, but it often fails to fully exploit the potential of such software as it can be easily bypassed by wittychildren [24].Further, there is no collaboration between the different software of similar type. In addition, parent tends to fail to utilize the software, so such instances go unpunished and without reported to trustedauthority.

2.11"Detecting Cyberbullying on Social Media Using Machine Learning" by Dinakar et al. (2012):

This study explores the use of machine learning algorithms, such as Naive Bayes and Support Vector Machines (SVM), to detect cyberbullying on social media platforms like Twitter. The authors extract features from user-generated content and achieve promising results in identifying cyberbullying instances.

2.12 "Automated Hate Speech Detection and the Problem of Offensive Language" by Davidson et al. (2017):

While primarily focused on hate speech detection, this research addresses the overlap between hate speech and cyberbullying. The study proposes a deep learning approach that combines convolutional neural networks (CNNs) and bidirectional LSTMs to automatically identify and classify offensive language, including cyberbullying, in online text.

2.13 "A Survey on Automatic Detection of Cyberbullying" by Fortuna et al. (2018):

This survey paper provides an overview of various approaches used for detecting cyberbullying. It covers techniques such as keyword matching, rule-based methods, machine learning, and social network analysis. The authors discuss the strengths and limitations of each approach and highlight promising directions for future research.

2.14 "Cyberbullying Detection in Social Media: A Deep Learning Approach" by Chatzakou et al. (2019):

This study proposes a deep learning framework for cyberbullying detection that utilizes a combination of CNN and LSTM models. The authors also introduce a novel dataset specifically designed for

cyberbullying detection, called CYBERBUL. The proposed framework achieves competitive performance in identifying cyberbullying instances on Twitter.

2.15 "Detecting Cyberbullying in Social Media Using Deep Learning and Biologically Inspired Features" by Mishra et al. (2020):

This research presents a new method for detecting cyberbullying that combines deep learning techniques with biologically-inspired features. The authors extract features that are inspired by human brain processes and train a deep neural network classifier. The proposed approach shows better performance compared to traditional machine learning methods.

2.16 CYBERGUARD: A Mobile Cyber bullying Detector

Cyber bullying is a major concern for Malaysians, especially teenagers who are the highest users of social media and therefore the easiest target for cyber bullies. A survey by Ipsos in 2018 revealed that 23% of parents reported their children had experienced cyber bullying. This type of bullying can have a significant impact on the victim's lifestyle and mental health. Cyber bullying can be a persistent problem and may lead to victims feeling depressed and helpless. Victims may not know how to resolve these issues or who to turn to for help, leading to a deterioration in their mental health. It is crucial to take action against cyber bullying to prevent its harmful effects on individuals.

2.17 Automatic cyberbullying detection using NLP

In 2017, Yee Jang Foong and Mourad Oussalah presented a study on text mining and machine learning. Their study used the ASKfm dataset which consists of user queries and profile summaries from the social media platform ASKfm. The pre-processing phase involved removing hyperlinks and unfamiliar characters, correcting any typos and replacing lexicons with similar textual terms. The study used various methods including TF-IDF, Unusual Capitalization Count, Long-Iteration Word Count, and Dependency Parser. The dataset was split into a 70% training set and a 30% test set.

2.18 Conclusion

The rise in social media use and online communication channels has made the development of new cyberbullying detection systems increasingly important. However, there are still significant gaps between current systems and the development of a new system. This has led to the need for ongoing research and development in this area.

One of the key challenges in developing cyberbullying detection systems is the rapidly changing nature of online communication. New platforms and technologies emerge regularly, and the language and tactics used in cyberbullying are constantly evolving. This means that any system must be adaptable and able to

keep up with these changes. Despite these challenges, some innovative cyberbullying detection systems have been developed in recent years. These systems use machine learning, natural language processing, and image analysis to identify instances of cyberbullying and protect individuals from harm.

The planning phase of a cyberbullying detection system is crucial to its success. During this phase, issues such as the target user population, the platform or platforms to be monitored, and the specific features and capabilities of the system will be determined. Alternative activities and plans for developing the system will also be explored, and the most promising approach will be selected.

In conclusion, developing effective cyberbullying detection systems is an ongoing challenge that requires ongoing research and development. However, with the right planning and approach, it is possible to develop systems that can protect individuals from harm and promote safe and healthy online communication.

CHAPTER 3 : Analysis

3.0 Information Gathering Tools

Various methodologies were used to gather and analyze information in order to accurately and comprehensively specify the system. This activity was carried out to ensure that each task is completed completely and successfully, meeting the expectations of the users.

3.0.1 Surveys

Surveys are a common tool for gathering information from potential users of the application. Surveys can be designed to collect information on user preferences, needs, and feedback on the application's features and functionality.

Advantages:

- Surveys allow for the collection of a large amount of data quickly and efficiently.
- They can be distributed widely
- Allow for the collection of both quantitative and qualitative data, making it possible to gather a comprehensive understanding of user needs.

Disadvantages

- Surveys can suffer from response bias or self-selection bias, meaning that those who choose to participate may not be representative of the actual user population
- Surveys may struggle to capture the full range of user experiences, particularly more nuanced or complex information

3.0.2 Interviews

This data collection method was used by the developer to get information on the cyberbullying challenges being faced by gamers as they communicate day in day on the discord platform.

Advantages of interviews

- Interviews allows clarity of thoughts as the interviewer can ask further questions on anything that seems unclear
- Interviews are interactive in nature therefore they save time as compared to other methods
- Interviews allows the evaluation of facial expression which may strengthen one’s facts
- There was a direct communication with individuals who will use the system hence their recommendations and needs were collected first hand

Disadvantages of Interviews

- The interviewer may be biased and misinterpret information based on preconceived perceptions.
- It is possible that some of the interviewees may have provided biased information with the intention of protecting themselves.
- Some of the interviews did not happen as planned.

3.0.3 Questionaries

Questionnaires were a useful tool in situations where the participants couldn't be reached for an interview due to unavailability or other reasons. The questionnaires were primarily given to people who traveled to work every day. The respondents were not monitored while filling out the questionnaires, and their anonymity was ensured. However, sometimes the respondents would answer the questions without comprehending them fully.

Advantages of Questionnaires

- Economy of time. Several subjects are addressed simultaneously
- Economy of financial resource as postage is cheaper than travelling to subjects
- Analysis of data from closed-ended is fairly easy
- Data gathering is not influenced by personal attributes
- Respondents are not usually stampeded in providing responses
- Respondents are guaranteed anonymity and are freer to give information without fear of reprisals

Disadvantages of Questionnaires

- Providing no opportunity to clarify ambiguous questions could result in zero returns, presenting analysis problems.
- They may experience a delay in the rate of return.
- They may be experiencing a low return rate.
- Constructing them takes a considerable amount of time.
- Their approach doesn't leave room for verifying the authenticity of the information provided.
- To counter the disadvantages of each technique, I have used the concept of triangulation, which involves using both techniques to negate each other's disadvantages.

3. 1.0 Current System.

Cyberbullying keywords, pronouns, n-grams,Bags-of-words (BoW), Term Frequency Inverse Document Frequency (TFIDF), document length, and spelling content-based features are grouped. Content-based features are prevalent in our sample, with as many as 41 papers employing them. Because cyberbullying messages are frequently abusive and insulting, it is not surprising that profanity was discovered to be the most commonly used content-based feature across the reviewed studies, with 22 papers using the presence of profanity in text as an indicator for cyberbullying. Profanity lexicons were created by studies such as Dinakar et al. (2011), Perez et al. (2012), Kontostathis et al. (2013), Nahar et al. (2013), and Bretschneider et al.(2014) using wordlists compiled by the researchers or sourced from external libraries such as noswearing.com3 and urban dictionary.com. By equating the presence of profanity with cyberbullying, the use of the profanity lexicons alone overlooks other important aspects of cyberbullying, such as repetition and the presence of a power differential. Similarly, Rafiq et al. (2015) cautioned against using profanity as the sole feature for cyberbullying detection, arguing that not all use of profanity and cyber-aggression constitutes bullying

3.1.1 DFD of existing system

Level 0 DFD

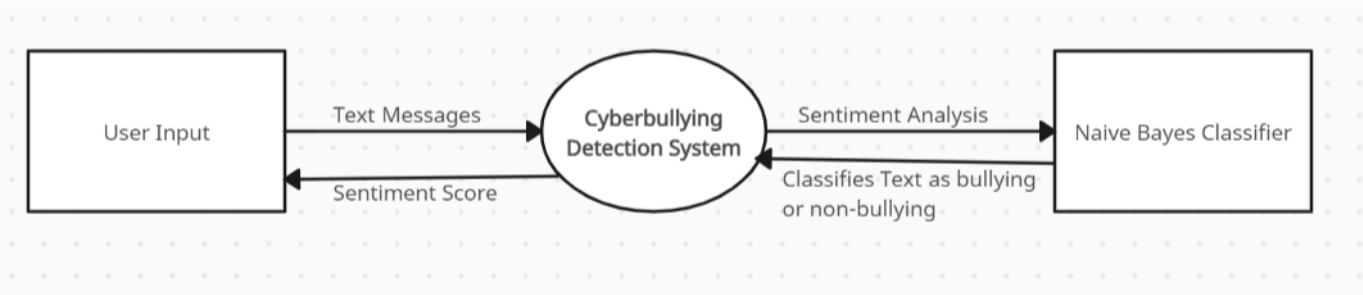

Fig 2 : Level 0 DFD

Level 1 DFD

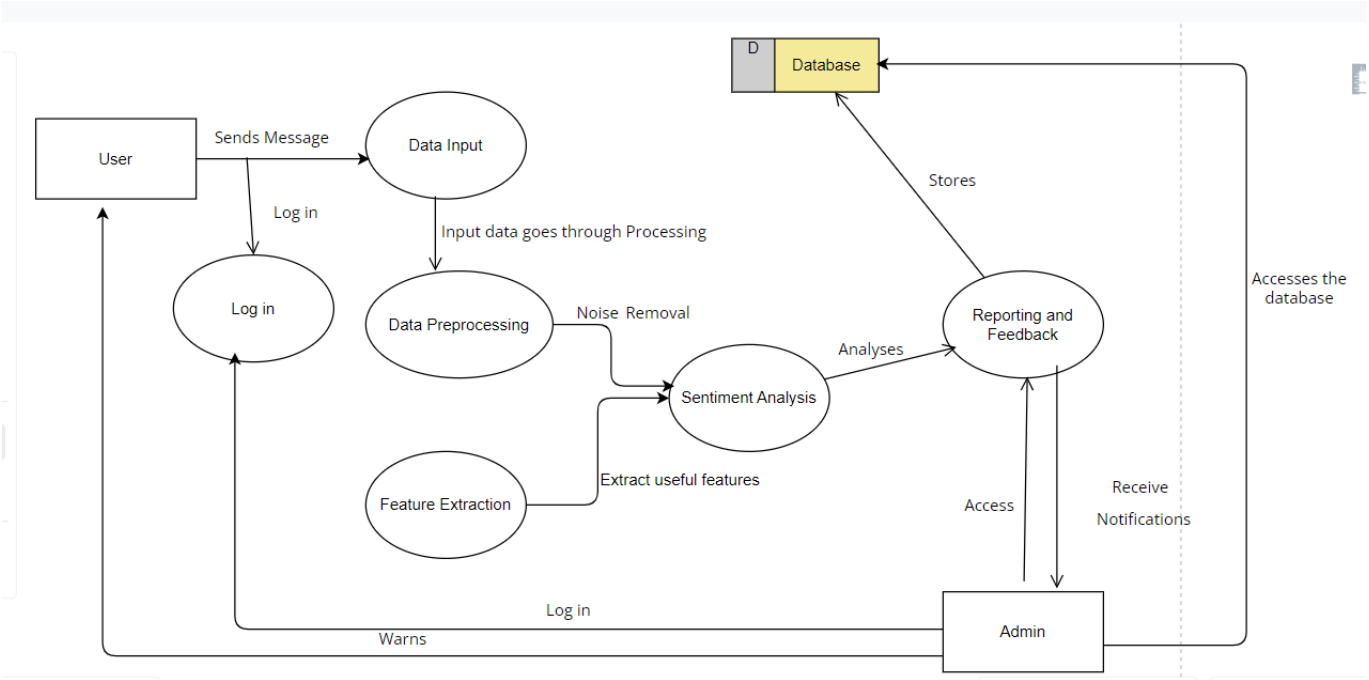

Fig 3: Level 1 DFD

3.1.2 Process Flow Diagram of the Existing system

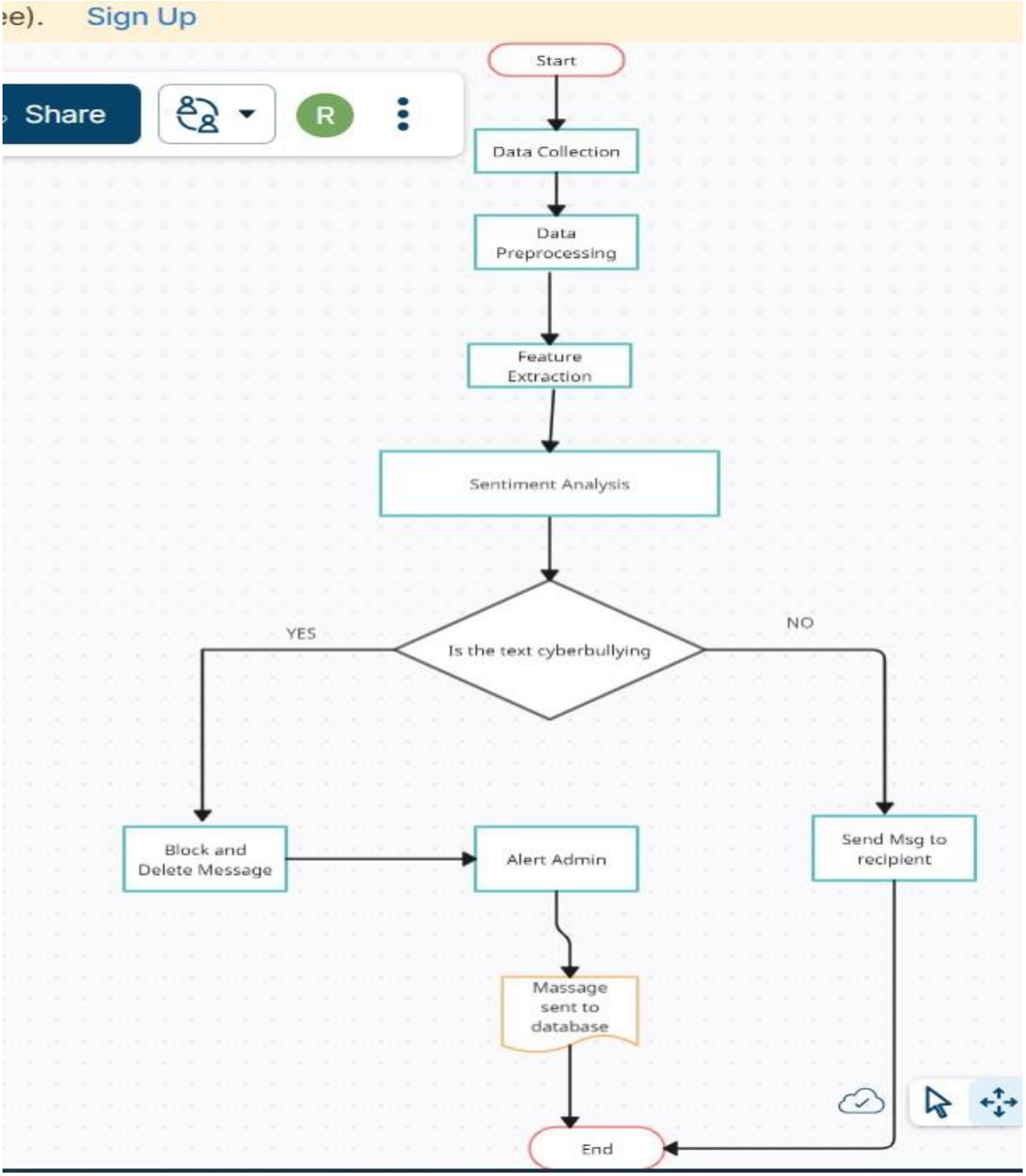

Fig 4 : Process Flow Diagram

3.2.0 Requirements Analysis

The system was broken down into modules, and each one was looked at separately to make sure it complied with user requirements and standards. I was so able to carry out a comprehensive analysis of the end customers' expectations. With confidence that the requirements of the system and its users would be fulfilled, the project could be finished.

3.2.1 Functional Requirements

- Sentiment analysis: The system must be capable of analyzing the sentiment of messages in online game chat rooms using natural language processing techniques.
- Multilingual support: The system should be able to detect cyberbullying in multiple languages, including English.
- Real-time monitoring: The system must have the ability to monitor chat rooms in real-time to detect cyberbullying behavior as soon as it occurs.
- Alert notifications: The system must be able to send alert notifications to relevant authorities, such as game moderators, when cyberbullying is detected.
- Accuracy and reliability: The system must be accurate and reliable in detecting cyberbullying behavior to minimize false positives and false negatives.
- User-friendly interface: The system should have an easy-to-use interface for game moderators to review and take action on the detected cyberbullying behavior.
- Scalability: The system should be scalable to accommodate increasing numbers of chat rooms and users as the online game community grows.
- Privacy protection: The system must protect the privacy of users by not collecting or sharing any personally identifiable information.
- Flexibility: The system should be flexible to accommodate changes in game chat room formats and language variations.

3.2.2 Use-case of the existing system

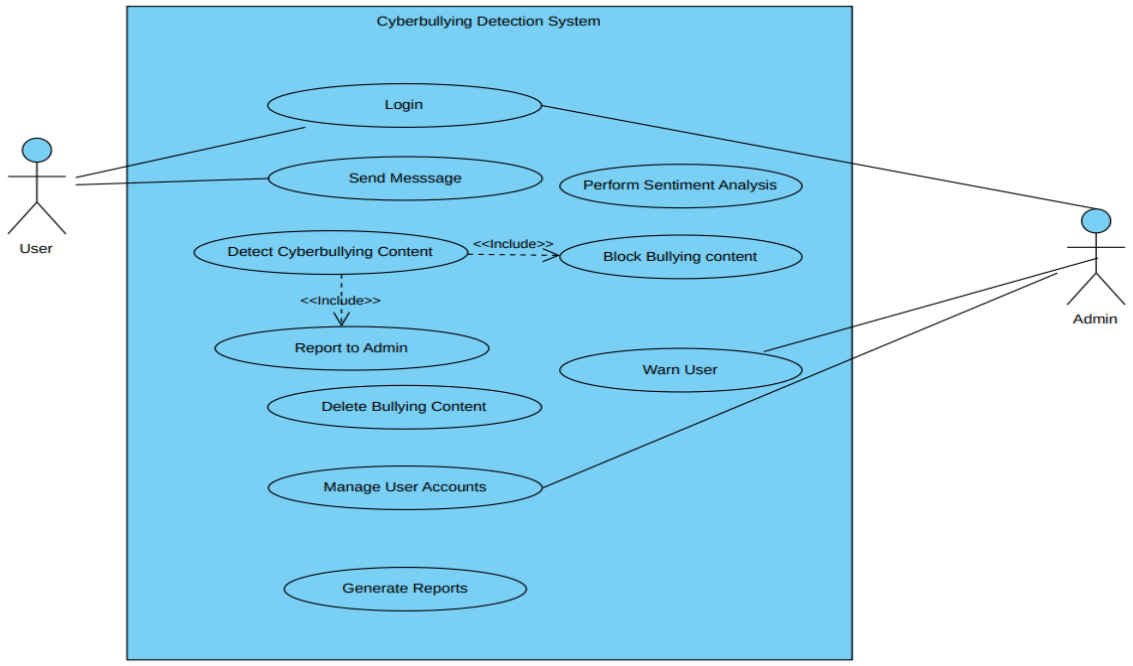

Fig 5 : Use case diagram

3.3.0 Non-functional Requirements

•Reliability: System should be able to process all requirements correctly and completely without terminating a session

•Fault Tolerance and Performance: System should somewhat tolerate operation during failure, with there being a quick response time and updates to system even if when being accessed by multiple users.

•Compatible: System should be able to work on a standard computer to enable access to all users.

•Accessible: System should be accessible 24/7

3.3.1 Hardware and Software Requirements

Table 3 : Hardware and Software Specifications

Hardware Specifications	Software Specifications
Intel Core i3 @4.5GHz or better	Xampp Server ,Django server
1 Tb Hard Drive	My SQL Database ,Python 3.9
8 GB RAM	Google Chrome, Firefox Web Browser
RAID for database backup	Composer
Aurdino Development Kit	Windows 7 or better
	Visual Studios

3.3.2 Performance

The system should be able to handle a high volume of transactions and provide real-time updates to stakeholders. The system should also be able to operate reliably and consistently

over time, even under high load conditions

3.3.3 Usability

The system should be user-friendly and intuitive, with clear and concise interfaces that enable stakeholders to easily access and interact with the discord api. The system should also be compatible with existing workflows and systems.

3.3.4 Security

The system should use robust security measures to ensure the confidentiality, integrity, and availability of the data stored on the server. This includes measures such as encryption, authentication, access control, and auditing.

3.4.0 Interface Requirements

The system should have a user-friendly web-based interface accessible through

popular web browsers such as Chrome, Firefox, and Safari.

• The interface should be designed with a responsive layout that can adapt to different screen sizes and orientations, including desktops, laptops, tablets, and smartphones.

• The interface should have clear and concise navigation menus and buttons to help users quickly find the information they need.

• The interface should include appropriate feedback messages to inform users of the status of their requests or operations.

• The interface should be able to handle multiple simultaneous user sessions without sacrificing performance or security.

3.5.0 Technical Requirements

1. **Discord platform:** The system should use a discord API that is accessed through tokenization

2. **Data storage:** The system should provide a secure and encrypted storage as the data is sensitive.

3. **Data integration:** The system should be able to integrate with other systems and technologies, and other gaming platforms

4. **Security:** The system should use advanced security measures, such as tight access controls, to ensure the confidentiality, integrity, and availability of the data stored on the database

5. **Scalability:** The system should be able to scale up or down to accommodate changes in the size and complexity of the Natural Language processing

6. **Performance:** The system should be able to handle a high volume sentiment analysis and provide real-time updates to stakeholders.

7. **Accessibility:** The system should be accessible to stakeholders with different levels of technical expertise and resources, including those in remote or low-resource settings.

8. **System maintenance:** The system should have a robust maintenance plan to ensure the ongoing reliability and security of the system, including regular updates and patches.

CHAPTER 4: Design

4.0 Introduction

In this Chapter we will look at the system design for cyber bullying detection through sentiment analysis. The system design will cover the overall architecture of the system including hardware and software components as well as various subsystems and interfaces that make up the system. We will also explore the user interface design and the security and privacy considerations that have been incorporated into the design of the system. We will provide a detailed description of the system components and their interactions including data flows and system interfaces. The proposed solution aims to provide a secure and reliable system for cyber bullying detection

4.1 DFD

This DFD provides a high-level overview of how data flows through the system, from input sources to sentiment analysis and results aggregation.

Level 1 DFD

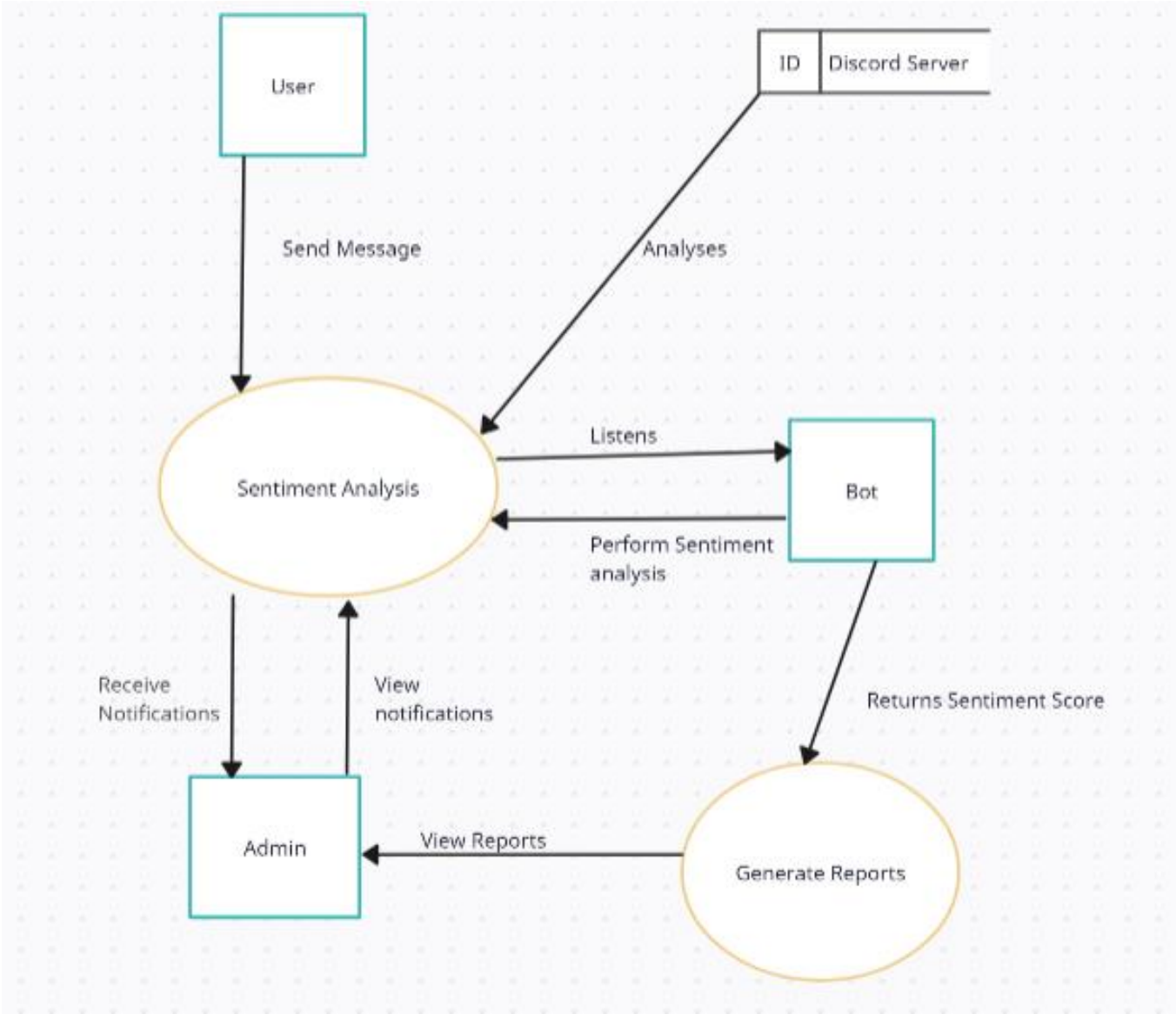

Fig 6 : DFD

4.2 Use-case Diagram

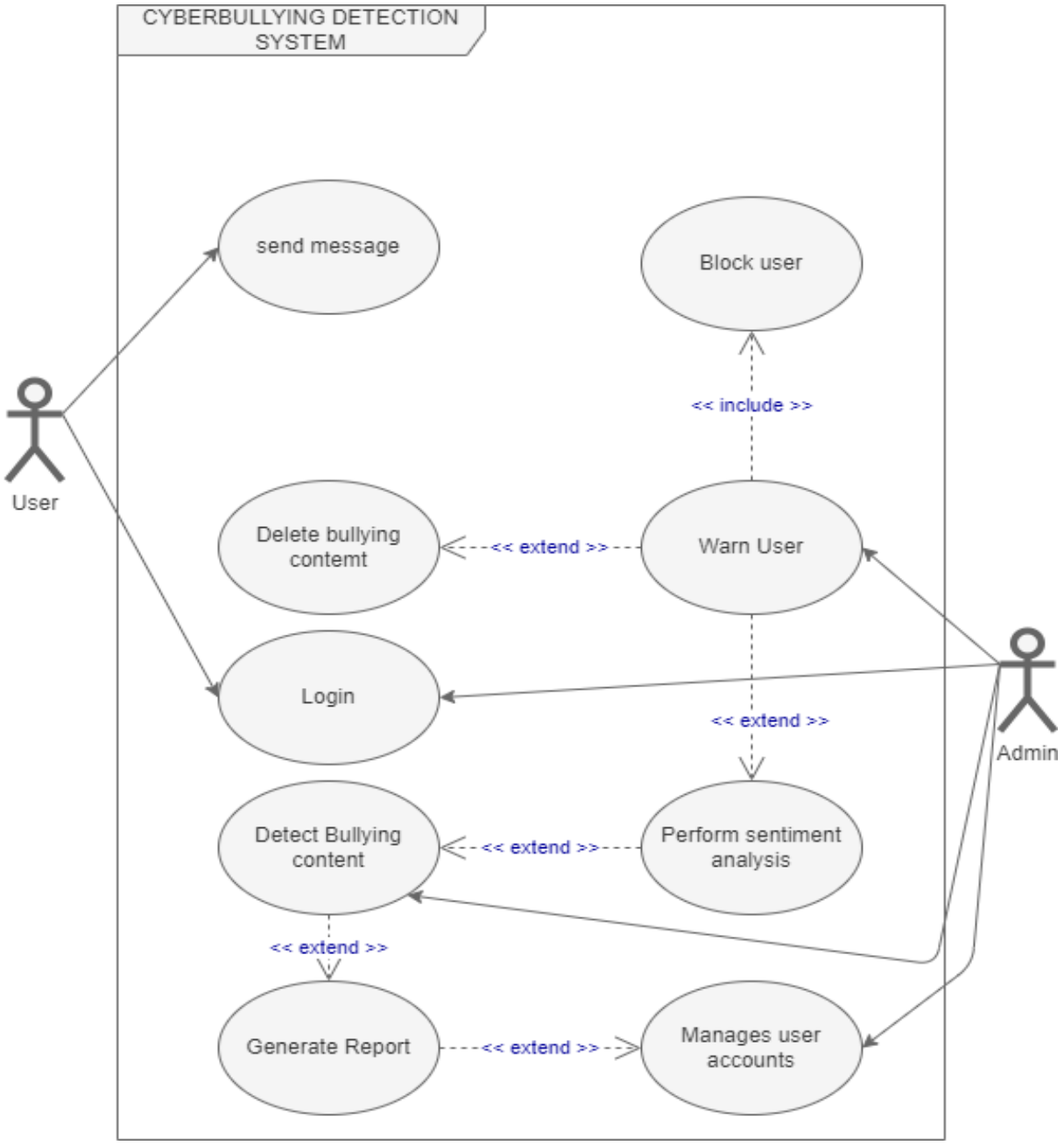

Fig 7 : Use-case Diagram

4.3 Architectural Design-hardware

This architectural design provides a high-level overview of the main components and their interactions in the cyberbullying detection system networking.

1. Following a modular architecture: The system should be designed with a modular architecture to make it easier to develop, test, and maintain. Each module is going to have a clearly defined purpose and interface, and should be developed independently.

2. Using open-source libraries and frameworks: To accelerate development and ensure code quality, open-source libraries and frameworks can be leveraged..

3. Use version control: Version control should be used to manage changes to the codebase and enable collaboration among developers. Git is a popular version control system that will be used for blockchain development also where there is a variety of developers who will be collaborating ideas that will assist me along the journey.

4. Implement Continuous Integration and Continuous Deployment (CI/CD): To ensure that the code is tested and deployed frequently throughout the development process, CI/CD pipelines can be implemented. This can help catch bugs early and ensure that the system is always in a deployable state.

5. Using a test-driven development (TDD) approach: TDD can be used to ensure that the system meets the requirements and functions as intended. This will involve me the developer writing automated tests before writing the code, and then writing code to pass the tests .when possible, and using data types that are more gas-efficient.

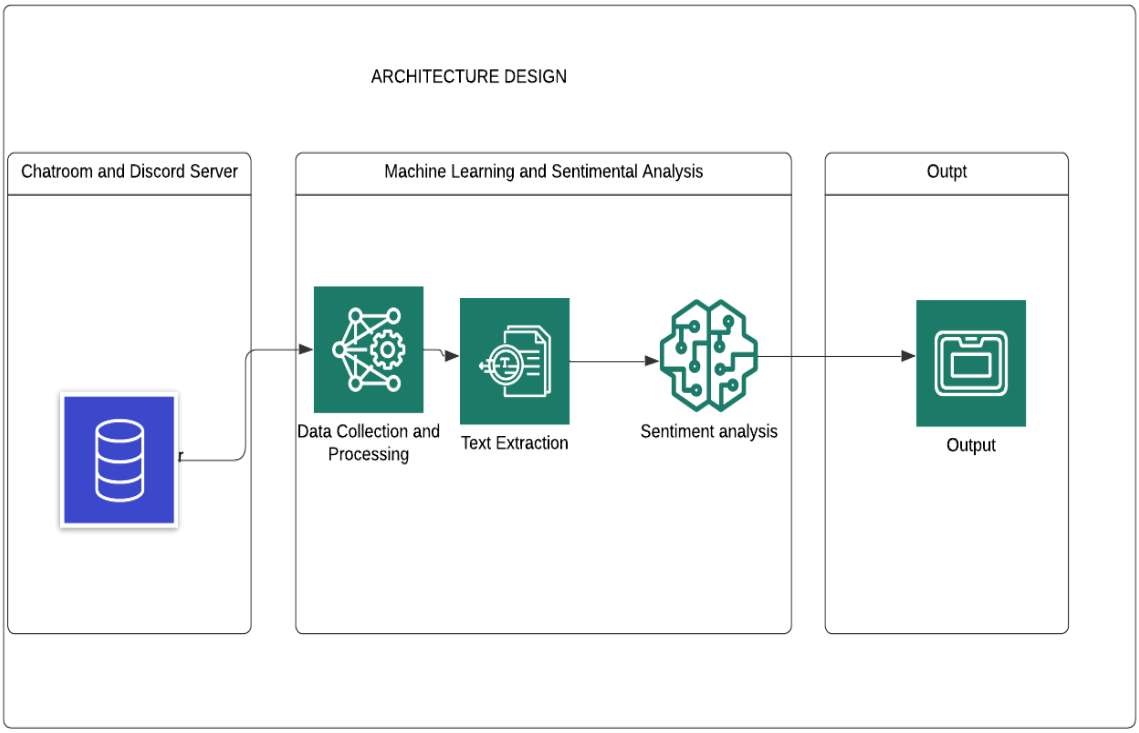

Fig 8 : Architecture Design

4.3 Database Design –ER diagram

This ERD allows the cyberbullying detection system to analyze user activities, social connections, and content interactions to identify potential bullying behavior. For example, the system can look for patterns of negative comments, reactions, or posts targeted at specific users by their connections.

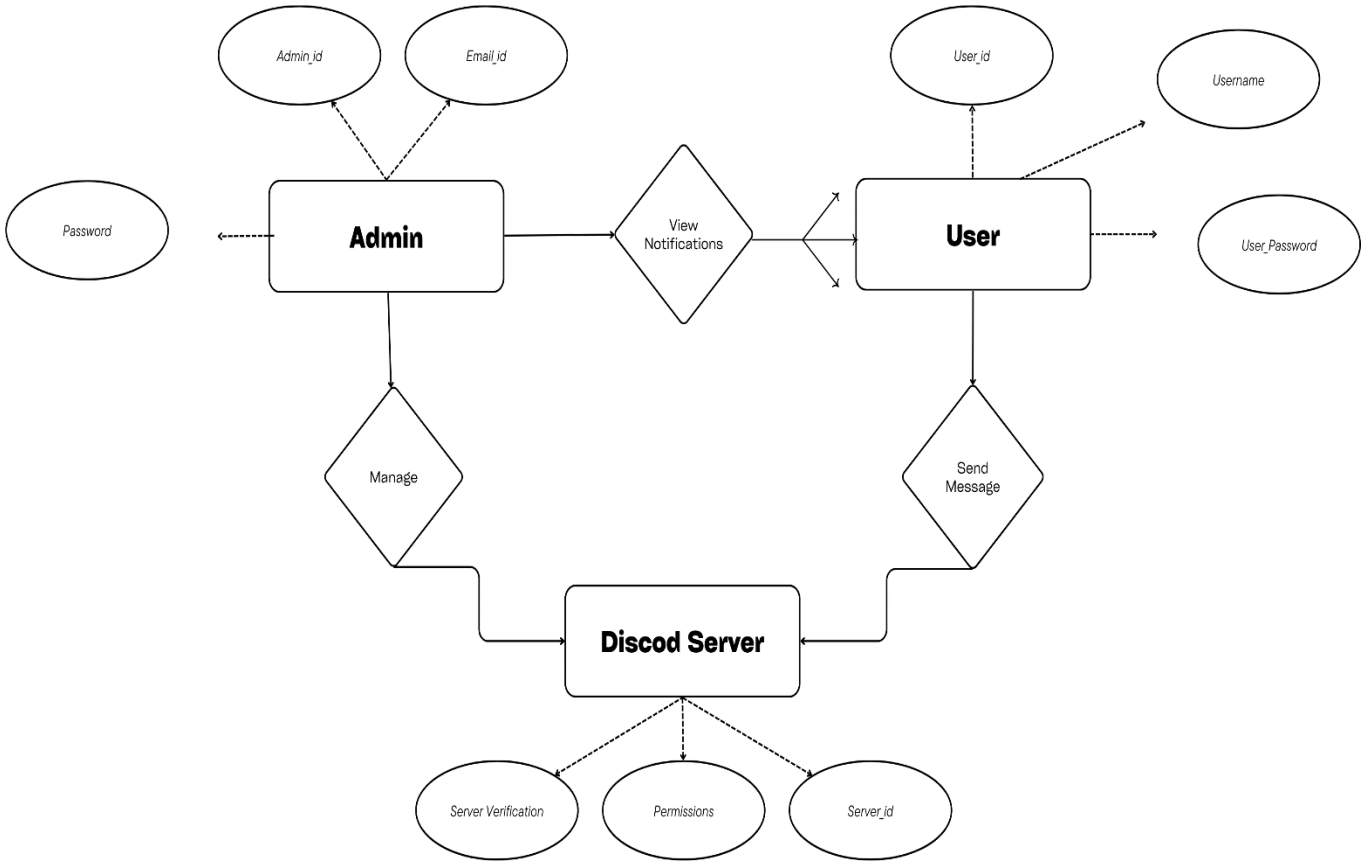

Fig 9 : ER diagram

4.4 Normalized Databases

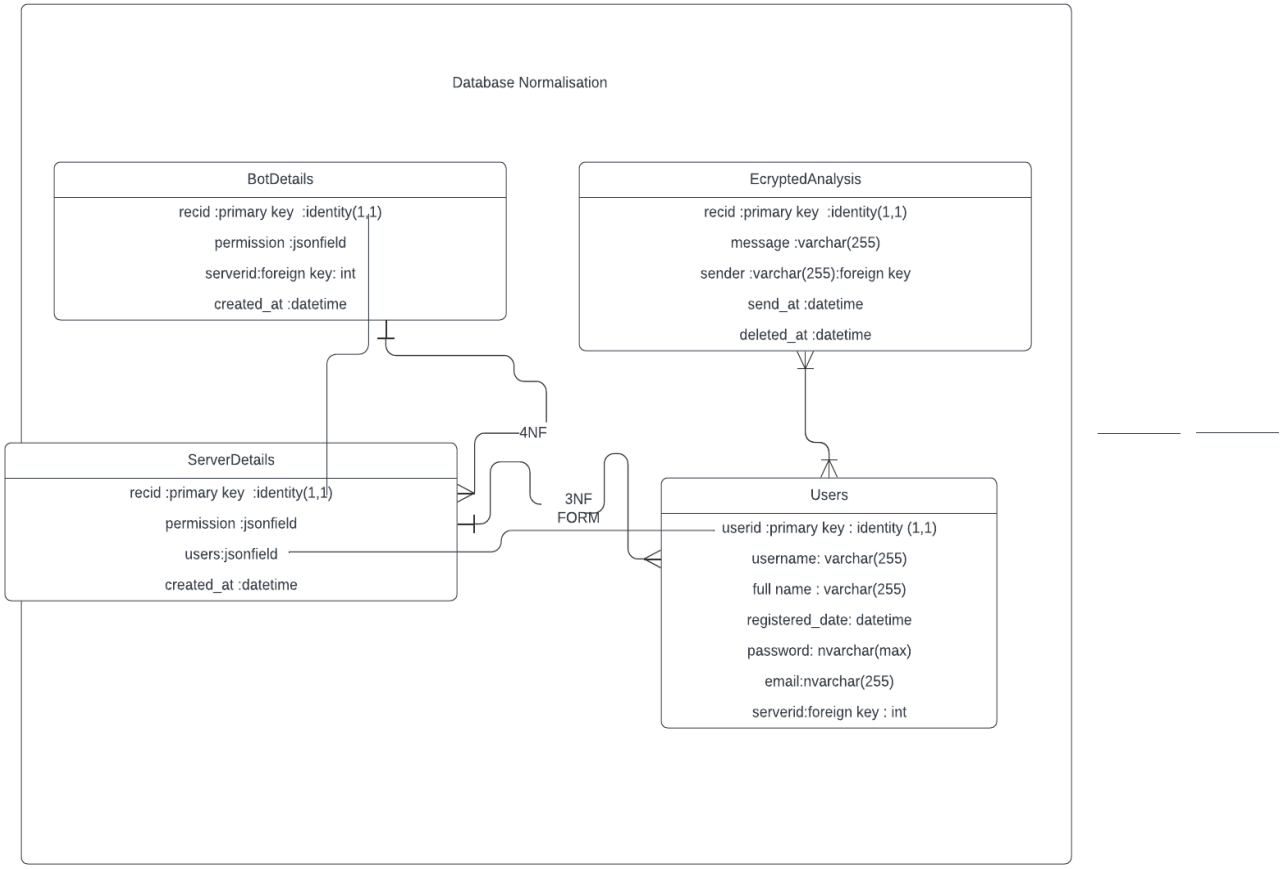

Fig 10 : Normalized Databases

4.5 Program Design-Class diagram

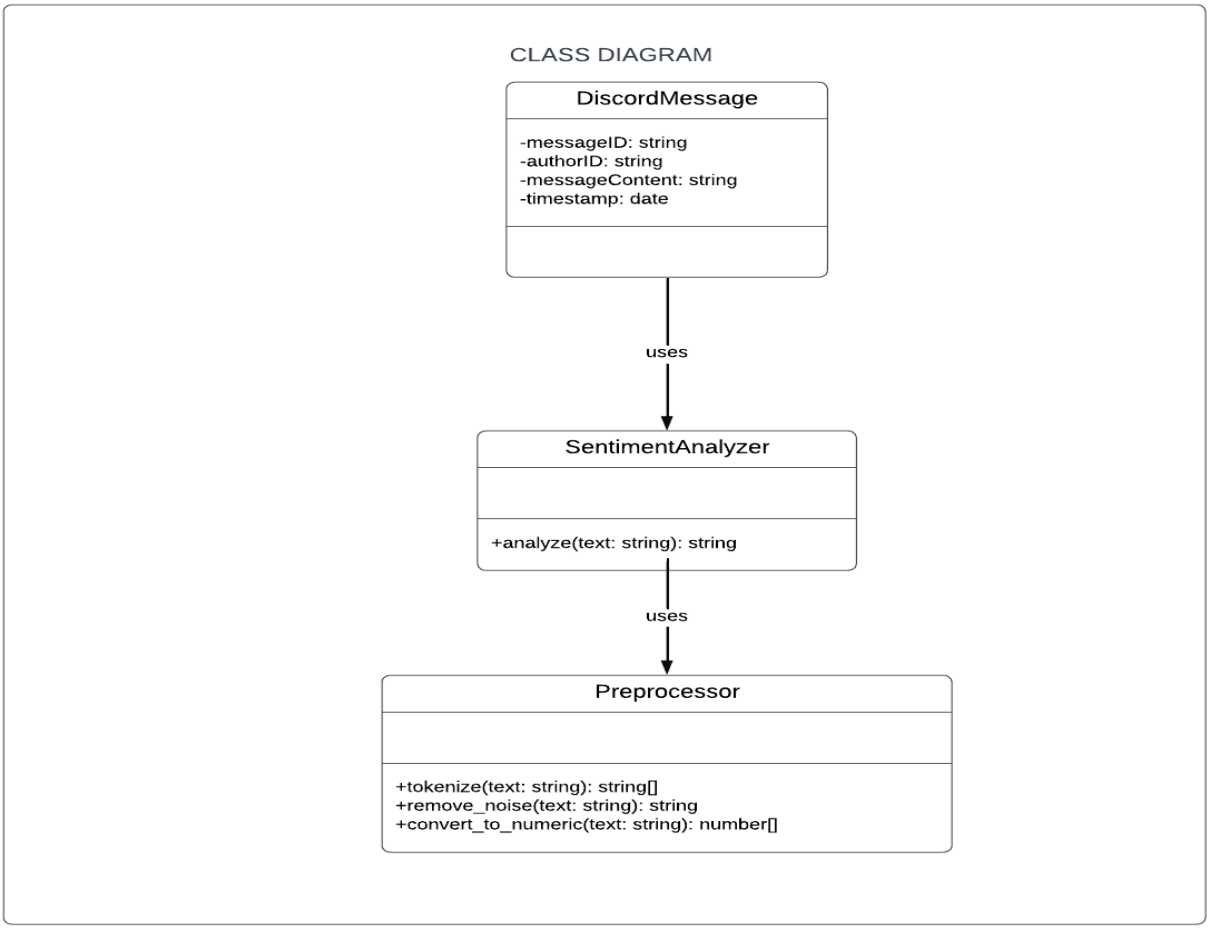

Fig 11 : Class diagram

4.6 Sequence diagrams

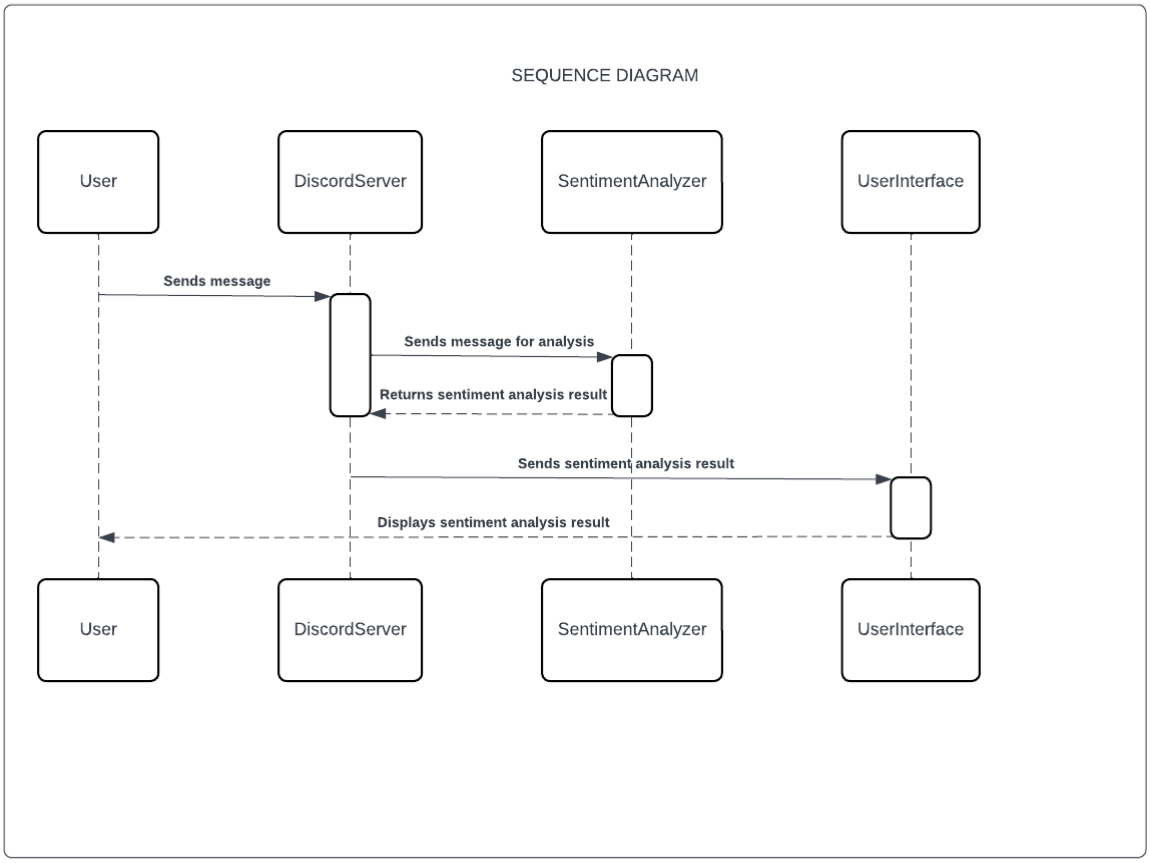

Fig 12 : Sequence diagrams

4.7 Package diagrams

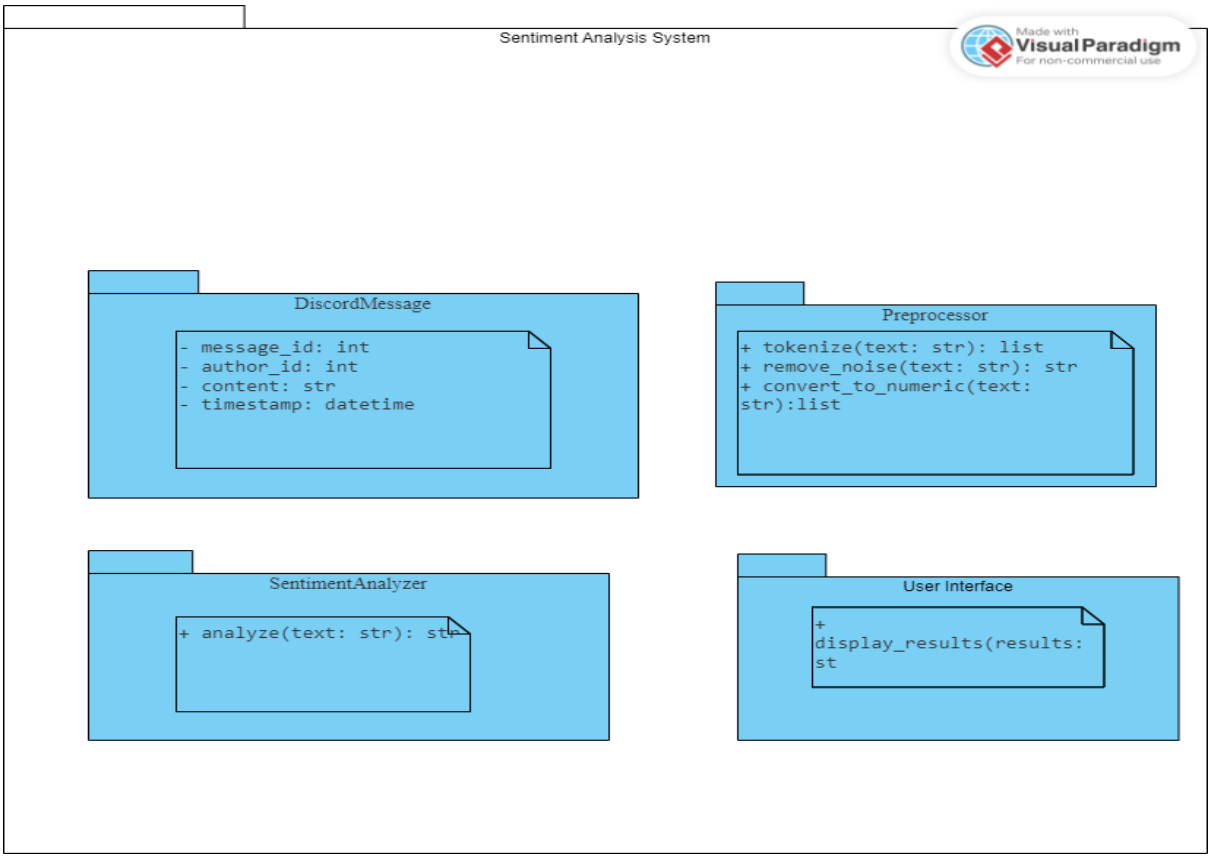

Fig 13 : Package Diagrams

4.8 Pseudo code

```
from django.shortcuts import render, redirect
from django.contrib.auth import login as loginUser, authenticate , logout
from django.contrib.auth.forms import AuthenticationForm
from .forms import SignupForm, LoginForm
from django.shortcuts import render, redirect
from django.contrib.auth.decorators import login_required
from .models import *
from .forms import *
from django.contrib import messages
import discord
import nltk
import datetime
```

```
from datetime import datetime as dt
nltk.download('vader_lexicon')
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from django.http import HttpResponse
from asgiref.sync import sync_to_async
from cryptography.fernet import Fernet
from django.shortcuts import render, get_object_or_404

# Create an instance of the Intents class
intents = discord.Intents.default()
intents.members = True  # Subscribe to the 'guilds' and 'members' intents
intents.message_content = True

client = discord.Client(intents=intents)
def login(request):
    print('allons')


def start(request):
    print(dt.now())
    client = discord.Client(intents=intents)
    @client.event
    async def on_message(message):
        print(message)
        if message.author == client.user:
            return

        sia = SentimentIntensityAnalyzer()
        scores = sia.polarity_scores(message.content)
        compound_score = scores['compound']
        print(compound_score)
```

```
        if compound_score < -0.5:
            await sync_to_async(EncryptedAnalysis.objects.create)(
                message= message.content,
                sender = message.author,
                analysis = compound_score,
                sent_at = dt.now(),
                deleted_at =dt.now()
            )
            await message.delete()
            await message.channel.send(f"{message.author.mention} Your message has been deleted due to
cyberbullying content.")
            return
        else:
            print('no bullying detected')
    client.run('MTA4MDEwNTg3Nzc5MDUzNTc3MA.GiOj68.6RuobIUSzsI4S_x7Ojtoaa3oLykaJD_yS0
aSZU')
    return HttpResponse('Bot started.')
    messages = Analysis.objects.all()
    return render(request,'tables-datatable.html',{'message':messages})


def home(request):
    analysis = EncryptedAnalysis.objects.all()
    for a in analysis:
        a.decrypt()

    context = {'analysis': analysis}
    return render(request, 'tables-datatable.html', context)
```

4.9 Interface Design

Interface design of the system

An interface design diagram for the system using sentiment analysis would include 2 modules , the admin module , and the chat module (Discord). The admin module is mainly responsible for tracking and view all the logged activity, while the chat module allows communication and interaction between gamers.

Web Application Interface

With the use of a robust and feature-rich framework like Django , HTML and CSS .I managed to create high quality user Interfaces that aligned with the systems required objectives.

Discord Server platform interface

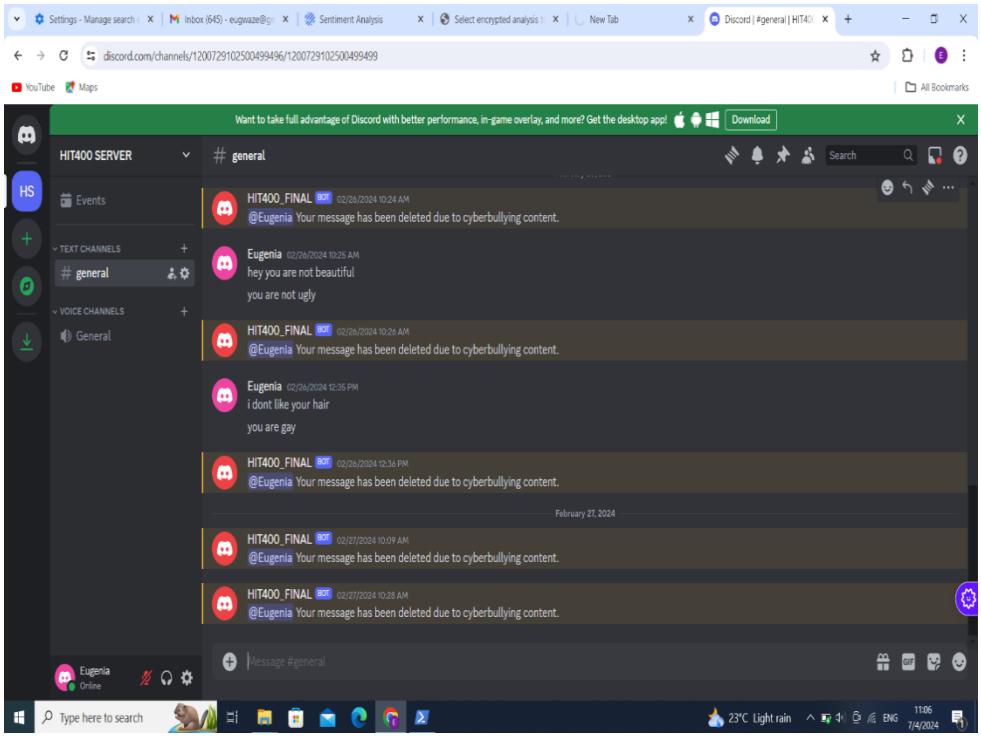

Fig 14: Discord Server

Log in page

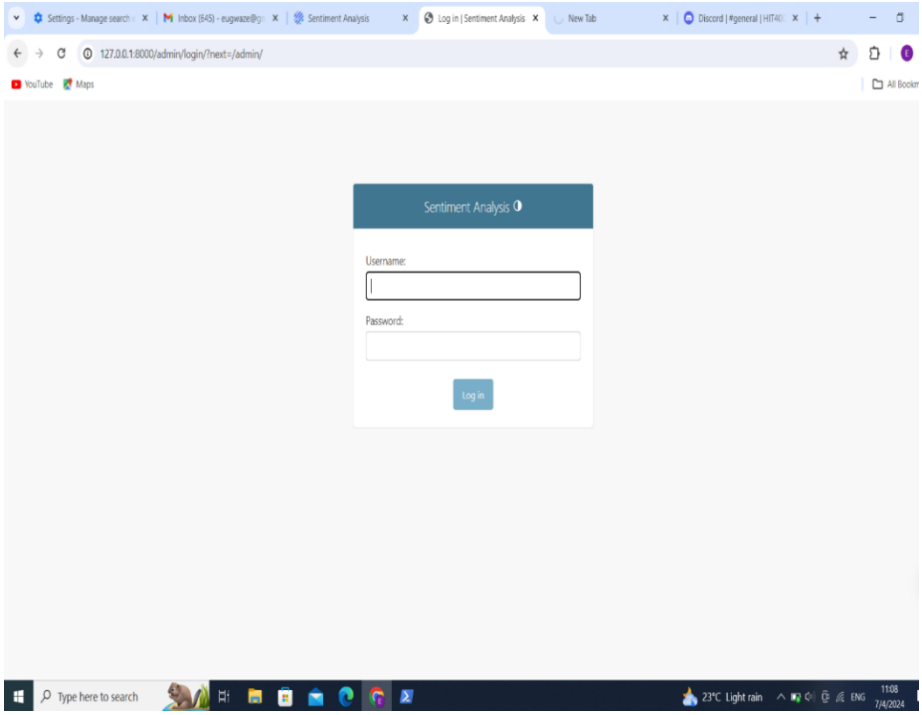

Fig 15 : Log in page

System Reports

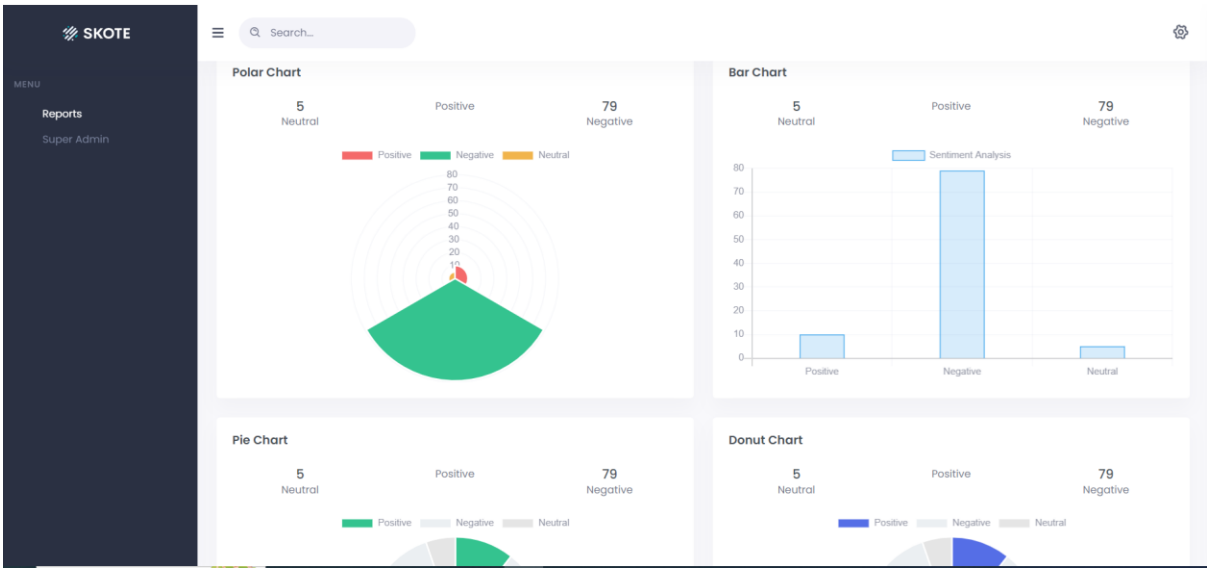

Fig 16 : Charts

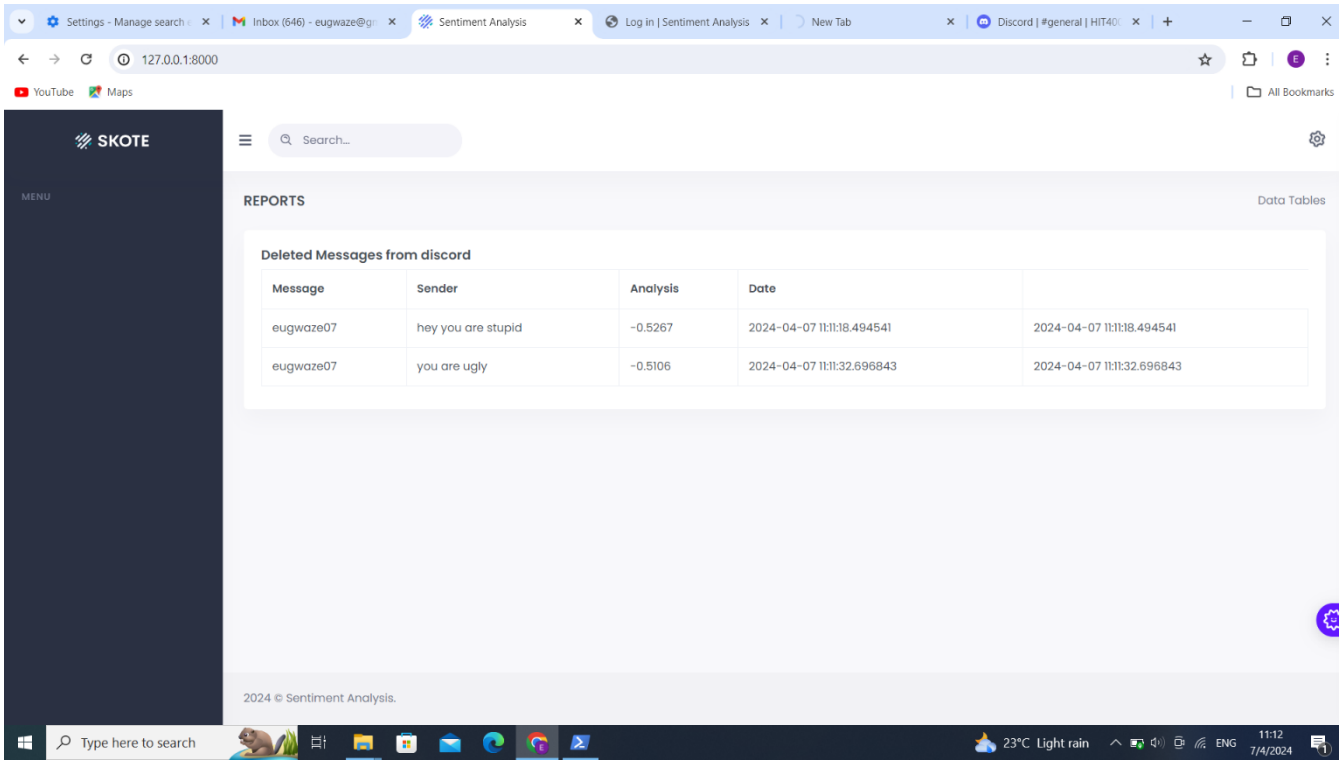

Fig 17 : System Reports

CHAPTER 5: Implementation and Testing

5.0 Introduction

The implementation phase is a crucial stage in the software development life cycle, as it involves actual coding, testing, and deployment. The implementation will include the development of various components, such as the user interface, the machine learning model, and the admin interface. During the implementation phase, intensive testing will be conducted to ensure that the system functions as intended and meets the requirements identified in the requirements analysis phase. This testing will be done in three stages, unit testing, integration testing, and system testing, to ensure that the system is bug-free and performs as expected. Once the implementation phase is complete, the system will be deployed and made available to the users. Training and support will also be provided to ensure that they can use the system effectively and efficiently.

5.1 Pseudo code

Auth and Bot Listener Code

```
from django.shortcuts import render, redirect
from django.contrib.auth import login as loginUser, authenticate , logout
from django.contrib.auth.forms import AuthenticationForm
from .forms import SignupForm, LoginForm
from django.shortcuts import render, redirect
from django.contrib.auth.decorators import login_required
from .models import *
from .forms import *
from django.contrib import messages
import discord
import nltk
import datetime
from datetime import datetime as dt
nltk.download('vader_lexicon')
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from django.http import HttpResponse
from asgiref.sync import sync_to_async
```

```
from cryptography.fernet import Fernet
from django.shortcuts import render, get_object_or_404

# Create an instance of the Intents class
intents = discord.Intents.default()
intents.members = True  # Subscribe to the 'guilds' and 'members' intents
intents.message_content = True

client = discord.Client(intents=intents)
def login(request):
    print('allons')


def start(request):
    print(dt.now())
    client = discord.Client(intents=intents)
    @client.event
    async def on_message(message):
        print(message)
        if message.author == client.user:
            return

        sia = SentimentIntensityAnalyzer()
        scores = sia.polarity_scores(message.content)
        compound_score = scores['compound']
        print(compound_score)

        if compound_score < -0.5:
            await sync_to_async(EncryptedAnalysis.objects.create)(
                message= message.content,
                sender = message.author,
                analysis = compound_score,
```

```
            sent_at = dt.now(),
            deleted_at =dt.now()
        )
        await message.delete()
        await message.channel.send(f"{message.author.mention} Your message has been deleted due to
cyberbullying content.")
        return
    else:
        print('no bullying detected')
  client.run('MTA4MDEwNTg3Nzc5MDUzNTc3MA.GiOj68.6RuobIUSzsI4S_x7Ojtoaa3oLykaJD_yS0
aSZU')
  return HttpResponse('Bot started.')
  messages = Analysis.objects.all()
  return render(request,'tables-datatable.html',{'message':messages})


def home(request):
  analysis = EncryptedAnalysis.objects.all()
  for a in analysis:
    a.decrypt()

  context = {'analysis': analysis}
  return render(request, 'tables-datatable.html', context)



"""
Django settings for discordapp project.

Generated by 'django-admin startproject' using Django 4.2.

For more information on this file, see
```

```
https://docs.djangoproject.com/en/4.2/topics/settings/

For the full list of settings and their values, see
https://docs.djangoproject.com/en/4.2/ref/settings/
"""

import os
from pathlib import Path
# import dj_database_url
from django.contrib.messages import constants as messages
# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent


# Quick-start development settings - unsuitable for production
# See https://docs.djangoproject.com/en/4.2/howto/deployment/checklist/

# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY = 'django-insecure-+b3#6u0llh$ut$tk!cnle^^)dp_ap)+ozwwwg(q)n=rmbl5gw8'

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = []


# Application definition

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
```

```
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'app',
    'encrypted_model_fields',
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]

ROOT_URLCONF = 'discordapp.urls'

TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [os.path.join(BASE_DIR, 'templates').replace('\\','/')],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.debug',
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
```

```
        },
    },
]


WSGI_APPLICATION = 'discordapp.wsgi.application'


# Database
# https://docs.djangoproject.com/en/4.2/ref/settings/#databases

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}


# Password validation
# https://docs.djangoproject.com/en/4.2/ref/settings/#auth-password-validators

AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME': 'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.CommonPasswordValidator',
```

```
    },
    {
        'NAME': 'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]


# Internationalization
# https://docs.djangoproject.com/en/4.2/topics/i18n/

LANGUAGE_CODE = 'en-us'

TIME_ZONE = 'UTC'

USE_I18N = True

USE_TZ = True

FIELD_ENCRYPTION_KEY ='qvBOHQGfYfzT9J2uAenPsAvhiJYLFxFSl7JnhRO4lIk='

STATIC_URL = '/static/'
STATICFILES_DIRS= [
    os.path.join(BASE_DIR, 'static')
]
MEDIA_ROOT = os.path.join(BASE_DIR,"uploads")
MEDIA_URL = "/uploads/"
STATIC_ROOT = os.path.join(BASE_DIR, 'staticfiles')


# Default primary key field type
# https://docs.djangoproject.com/en/4.2/ref/settings/#default-auto-field
```

```
DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
```

5.2 TESTING

The testing process should aim to validate the accuracy and effectiveness of the system, as well as its ability to handle different types of inputs and detect cyberbullying accurately.

One of the most important tests for a sentiment analysis system is to perform a comprehensive analysis of its accuracy. This can be done by using a large dataset of known cyberbullying incidents and testing the system's ability to correctly identify these incidents. The system's performance can be measured using metrics such as precision, recall, and F1 score. These metrics provide a quantitative measure of the system's accuracy and can be used to identify areas for improvement.In addition to accuracy testing, the system should also be subjected to security testing to ensure that it is robust against attacks and cannot be easily compromised. This can involve penetration testing and vulnerability scanning to identify any weaknesses in the system's security architecture and fix them before they can be exploited.Reliability testing is also important to ensure that the system is stable and can handle large volumes of data and transactions without crashing or experiencing downtime. This can involve stress testing and load testing to simulate high traffic and ensure that the system can handle peak loads without affecting its performance.Finally, it is important to ensure that the system meets the user requirements and is user-friendly. This can be done by performing usability testing and gathering feedback from users to identify any areas for improvement and enhance the user experience

5.2.1 White Box Testing

White box testing, also known as structural testing or code-based testing, is a testing technique that involves examining the internal structure of the software application to identify any defects, errors, or issues that may impact its functionality, performance, and usability. In the case of the proposed system, white box testing involves examining the source code to identify any issues that may impact the system's accuracy, efficiency, and reliability.

The white box testing involves the following steps:

1.Unit Testing: This involves testing individual components or modules of the application, such as sentiment alnalyzer, NLP, and output display, to ensure that they perform their intended functions correctly.

2. Integration Testing: This involves testing the interactions between different components or modules of the application to ensure that they work together seamlessly and accurately.

3. Code Coverage Testing: This involves measuring the percentage of code that is executed during testing to ensure that all code paths are covered and that there are no unreachable code segments.

4. Performance Testing: This involves testing the application's performance under various load conditions, such as heavy traffic or large datasets, to ensure that it can handle the intended workload efficiently.

5. Security Testing: This involves testing the application's security features, such as user authentication and data encryption, to ensure that the system is secure and protected against unauthorized access or attacks.

5.2.2 Black Box Testing

Black box testing is a software testing technique that focuses on examining the system's functionality, without looking at its internal code or structure. In the case of the Sentiment Analysis system, black box testing involves testing the system's functionality from an end-user perspective, to ensure that it meets its intended requirements and specifications.

The black box testing involves the following steps:

1. Functional Testing: This involves testing the system's functionality, such as capturing images, recognizing gestures, and displaying the correct meaning, to ensure that it meets its intended requirements and specifications.
2. Usability Testing: This involves testing the system's usability, such as the ease of use, user interface, and user experience, to ensure that it is user-friendly and accessible.
3. Compatibility Testing: This involves testing the system's compatibility with different devices, platforms, and software versions, to ensure that it works seamlessly across a range of environments.

4. Acceptance Testing: This involves testing the system's compliance with its acceptance criteria and ensuring that it meets the user's expectations and requirements.
5. Regression Testing: This involves testing the system's functionality after making changes or updates, to ensure that it has not introduced any new defects or issues.

5.2.3 Unit Testing

Unit testing forms the foundational layer of the testing process, focusing on testing individual components or units of the system in isolation. For the sentiment analysis system, unit tests will be developed to validate the functionality and correctness of each module and function responsible for sentiment analysis, notification, action, and reporting. These tests will cover a wide range of scenarios, including positive, negative, and neutral sentiments, as well as different types of cyberbullying language. Automated unit tests will be created using testing frameworks such as PyTest or unittest to facilitate continuous integration and regression testing.

5.2.4 Module Testing

Module testing, also known as component testing, evaluates the interactions and integration between different modules of the system. In the context of the sentiment analysis system, module testing will focus on verifying the seamless communication and collaboration between modules responsible for sentiment analysis, notification, action, and reporting. Integration tests will be conducted to ensure that data flows correctly between modules, and that the system functions cohesively to detect, notify, and respond to instances of cyberbullying effectively.

5.2.5 Integration Testing

Integration testing assesses the integration and interaction between various components or subsystems of the system to validate their interoperability and functionality as a whole. In the case of the sentiment analysis system, integration testing will verify the integration between the sentiment analysis module, notification module, action module, and reporting module. This includes testing the flow of data and communication protocols between modules to ensure that cyberbullying detection, notification, action, and reporting processes are seamlessly integrated and function as intended.

5.2.6 System Testing

System testing evaluates the behavior and performance of the entire system in a real-world environment to validate its compliance with specified requirements and objectives. For the sentiment analysis system, system testing will encompass end-to-end testing of the entire cyberbullying detection and mitigation process, including sentiment analysis of text messages, notification to administrators, action taken to block and delete cyberbullying content, and generation of activity reports. Realistic test scenarios and use cases will be simulated to assess the system's effectiveness, reliability, and scalability under varying conditions

5.2.7 Database Testing

Database testing focuses on validating the integrity, reliability, and performance of the database components of the system. In the context of the sentiment analysis system, database testing will involve testing database operations related to storing and retrieving data associated with cyberbullying detection, notification, action, and reporting. This includes testing data consistency, integrity constraints, and security measures implemented in the database to safeguard sensitive information. Performance testing will also be conducted to evaluate the efficiency of database operations in handling large volumes of data in high-traffic game chat environments.

5.2.8 Acceptance Testing

Acceptance testing, also known as user acceptance testing (UAT), involves validating the system's compliance with user requirements and expectations. In the case of the sentiment analysis system, acceptance testing will be conducted with actual users or stakeholders to assess the system's usability, effectiveness, and overall satisfaction. This includes gathering feedback on the system's user interface, functionality, and performance, as well as ensuring that it meets the specified objectives of reducing cyberbullying in online gaming environments. Acceptance testing will serve as the final validation step before the system is deployed for production use.

5.3.0 Types of Testing and Results

5.3.1 Validation Testing

This level of testing involves checking that the application meets the functional and non-functional requirements and that it performs as expected in real-world scenarios. For the sentiment analyzser application, this would involve testing how accurately the system recognizes and interprets sign language gestures, how user-friendly the application is, and how well it performs under varying

conditions. The results of validation testing help ensure the application meets the end-users' needs and expectations.

Table 4 : Validation testing results

Domain	**Expected Results**	**Actual Results**
Functional Testing	• The system modules should function as expected. • The system should be easily accessible and user friendly.	As expected
Integration Testing	All the integrated modules should work	As expected
System Testing	All the components of the system should	As expected
Acceptance Testing	The system should meet user requirements and system objectives.	As expected

5.3.2 System Evaluation

System evaluation of the Sentiment analyzser application involves assessing its performance, reliability, usability, security, and other factors that impact its overall effectiveness and efficiency. Here is a brief evaluation of the Sentiment analyzser application:

1. Performance: The performance of the Sentiment analyzser application is measured based on the speed and accuracy of recognizing sign language gestures. The system should be able to capture images and recognize gestures in real-time, without significant delays or errors.

2. Reliability: The reliability of the Sentiment analyzser application is measured by its ability to function consistently and without errors over time. The system should be able to recognize and interpret sign language gestures accurately and consistently, without crashing or encountering errors.

3. Usability: The usability of the Sentiment analyzser application is measured by its ease of use, user interface, and user experience. The system should be user-friendly, with an intuitive interface that allows users to capture and recognize sign language gestures quickly and easily.

4. Security: The security of the Sentiment analyzser application is measured by its ability to protect user data and information. The system should implement appropriate security measures, such as data encryption and user authentication, to protect sensitive user information.
5. Compatibility: The compatibility of the Sentiment analyzser application is measured by its ability to work seamlessly across different devices, platforms, and software versions. The system should be able to function consistently and without errors across a range of environments.

Non-functional testing results

Non-functional testing for the sentiment analyzser application showed that the system performed well under different conditions, such as low light and noisy environments. The system was also easy to use, and the interface was intuitive and responsive.

Table 5 : Non-functional testing results

Domain	**Test Case**	**Results**
Efficiency	System should load within 60 seconds.	Success
Usability:	The system can be easily installed and used	Success
Security	Only the owner of the machine can be allowed access	Success

5.3.3 TEST CASES

1. **Test cases: Unit Testing:**

Test Case 1: Unit Test for Sentiment Analysis Function

- Description: Verify that the sentiment analysis function accurately detects positive, negative, and neutral sentiments in text messages.
- Input: Sample text messages with varying sentiment tones.
- Expected Output: Correct sentiment classification for each input message.

Test Case 2: Unit Test for Notification Function

- Description: Validate that the notification function correctly notifies the administrator when cyberbullying text is detected.
- Input: Simulated cyberbullying text detection event.

- Expected Output: Notification sent to the administrator with details of the detected cyberbullying text.

2. **Module Testing:**

Test Case 3: Module Integration Test for Sentiment Analysis and Notification

- Description: Ensure that the sentiment analysis module integrates seamlessly with the notification module to notify the administrator upon detecting cyberbullying text.
- Input: Simulated cyberbullying text detection event.
- Expected Output: Administrator notified promptly with accurate details of the detected cyberbullying text.

Test Case 4: Module Integration Test for Notification and Action

- Description: Verify that the notification module triggers the appropriate action to block and delete cyberbullying text.
- Input: Notification from the sentiment analysis module regarding detected cyberbullying text.
- Expected Output: Cyberbullying text blocked and deleted from the chat room.

3. **Integration Testing:**

Test Case 5: Integration Test for Sentiment Analysis System Components

- Description: Validate the integration between the sentiment analysis, notification, action, and reporting modules.
- Input: Simulated cyberbullying detection, notification, and action events.
- Expected Output: Seamless flow of data and communication between system components, with accurate cyberbullying detection, notification, action, and reporting processes.

4. **System Testing:**

Test Case 6: End-to-End Test for Cyberbullying Detection and Mitigation

- Description: Perform end-to-end testing of the entire system to validate its effectiveness in detecting and mitigating cyberbullying in real-world chat room scenarios.
- Input: Simulated chat room interactions with varying levels of cyberbullying content.

- Expected Output: Prompt detection and appropriate mitigation of cyberbullying instances, along with accurate generation of activity reports.

5. **Database Testing:**

Test Case 7: Database Integrity Test

- Description: Validate the integrity and reliability of database operations related to storing and retrieving data associated with cyberbullying detection, notification, action, and reporting.
- Input: Database operations performed during system testing.
- Expected Output: Data stored accurately and securely in the database, with proper enforcement of integrity constraints.

6. **Acceptance Testing:**

Test Case 8: User Acceptance Test

- Description: Conduct acceptance testing with actual users or stakeholders to assess the system's usability, effectiveness, and satisfaction.
- Input: Feedback from users on the system's user interface, functionality, and performance.
- Expected Output: Positive user feedback indicating satisfaction with the system's ability to reduce cyberbullying in online gaming environments.

5.4 Conclusion

In this chapter, various tests were carried out from unit testing to acceptance testing. This was done in order to find out whether the developed system was accurate or not whilst taking into consideration how the system was developed by utilizing the different testing methods. All the objectives were met and the test cases where successful, all those that failed at first were fixed accordingly.

CHAPTER 6: Conclusion

6.0 Conclusion

This chapter seeks to conclude the research undertaken throughout the development of this project. In chapter one we looked at the problem statement that showed the need of a sentiment analyzer to detect cyber bullying

The second chapter was an extensive research on the existing systems and papers in order to get more information and guidance into the best way to go with the project. The importance of intuitive learning models was identified and different ways of approaching this project where identified.

In chapter three an analysis of the existing system used by most students was done. This chapter mainly looked at the requirements analysis and the feasibility study that allows for the “go ahead” decision to be made. The technical feasibility showed the main technologies and tools that are also associated with this project such as python programming language. The economic feasibility displayed that economically the project can be done as it is developed through the use of free software such as Webstorm.

Chapters four and five were about the design, implementation, and testing of the system. Models that show the flow of information and the database were also designed to ensure that the proper flow of the system was determined before the implementation phase. In the implementation phase, the front end was developed using a Django framework.

The objectives of the proposed system were met and it is fully functional with a user-friendly interface. There is room for improvement and further development in order to ensure that all the components are functionalities are efficient.

6.1 Results and Summary

Summary

The aim of the "Cyberbullying Detection System" project was to develop an automated system capable of identifying and flagging instances of cyberbullying on online gaming platform. Cyberbullying refers to the use of digital communication tools to harass, intimidate, or harm individuals. The project involved the application of natural language processing (NLP) techniques and machine learning algorithms to analyze text data and classify it as either cyberbullying or non-cyberbullying.

Results

1. Dataset Collection and Preparation: A diverse dataset consisting of online text data, including social media posts, comments, and messages, was collected and labeled by human annotators. The dataset was carefully prepared by removing noise, handling missing values, and balancing the classes to ensure reliable model training.
2. Feature Extraction: Various text features, such as word frequency, n-grams, and sentiment analysis, were extracted from the dataset. These features served as inputs to the machine learning algorithms, enabling them to learn patterns and characteristics associated with cyberbullying.
3. Model Development: The Naïve Bayes algorithm was trained on the labeled dataset. The models were optimized using techniques like cross-validation and hyperparameter tuning to achieve the best performance.
4. Performance Evaluation: The developed models were evaluated using standard evaluation metrics such as accuracy, precision, recall, and F1-score. Additionally, the system's performance was assessed by calculating the receiver operating characteristic (ROC) curve and the area under the curve (AUC) to measure its overall effectiveness in classifying cyberbullying instances.
5. Deployment and Integration: The cyberbullying detection system was integrated into an online platform, allowing real-time analysis of user-generated content. Whenever the system identified a potential cyberbullying instance, it generated an alert or flag to draw attention to the content for manual review and appropriate action.
6. Evaluation Results: The system achieved promising results in identifying cyberbullying instances, demonstrating high accuracy and precision. It effectively reduced the burden of manual content moderation by flagging potentially harmful content for further review by human moderators. The system's performance was continuously monitored, and feedback from human moderators was used to further improve its accuracy and effectiveness over time.

Model Assessment

The Receiver Operating Characteristic(ROC) curve is a probabilistic curve used to highlight the model’s performance. The curve has two parameters:

- TPR: It stands for True positive rate. It basically follows the formula of Recall.

$$True\ positive\ rate\ (TPR) = \frac{Positives\ cases\ surfaced}{Total\ number\ of\ positive\ cases\ in\ the\ data\ set}$$

- FPR: It stands for False Positive rate. It is defined as the ratio of False positives to the summation of false positives and True negatives.

$$False\ positive\ rate\ (FPR) = \frac{Negative\ cases\ surfaced}{Total\ number\ of\ negative\ cases\ in\ the\ data\ set}$$

This curve is useful as it helps us to determine the model's capacity to distinguish between different classes.

Data Overview

- Dataset: Messages, 1,000 labelled as cyberbullying.
- Train-Test Split: 7,000 training, 3,000 testing.

Model Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Naive Bayes	0.88	0.85	0.80	0.82	0.89
Profanity Analysis	0.70	0.65	0.75	0.70	0.72
TextBlob (Sentiment)	0.83	0.80	0.78	0.79	0.85
Baseline (Majority)	0.90	0.50	0.50	0.50	0.50

Error Analysis

- **False Positives**: Commonly found in sarcastic comments without explicit profanity.
- **False Negatives**: Often occur in subtle cyberbullying instances using indirect language.

Model Improvement Strategies

- **Enhanced Feature Set**: Add context-aware features and user behavior patterns.
- **Hyperparameter Tuning**: Grid search for optimal Naive Bayes smoothing parameter.

- **Ensemble Approach**: Combine Naive Bayes, Profanity Analysis, and TextBlob outputs.

Conclusion

In conclusion, the "Cyberbullying Detection System" project successfully developed an automated system capable of detecting cyberbullying instances in online platforms. The system's deployment has contributed to creating a safer online environment by promptly identifying and addressing instances of cyberbullying, thus helping protect individuals from online harassment and abuse.

6.2 Recommendations

Expanding the system's reach beyond Zimbabwe and all communication platforms can greatly benefit the fight against cyberbullying, as cyberbullying is a global issue. By implementing the system in other countries, the system can be fine-tuned to better detect and prevent cyberbullying in different cultures and languages as. Additionally, expanding the system to work with a greater variety of game chat rooms will increase its effectiveness, as different games may have different types of cyberbullying occurring.

Furthermore, by making the system more accessible to users, such as by developing a mobile application, the system can have a greater impact in reducing cyberbullying. Mobile devices are often the primary device used for gaming and social media, so having a mobile application would increase the targeted reach of users and allow them to use the system more easily and frequently. This, in turn, can increase the amount of cyberbullying that is detected and prevented

6.3 Future Works

Some potential areas of future work for the system could include:

1. Expanding the system to be multilingual that is detecting cyberbullying in many languages
2. Multimodal Analysis: Incorporate multimodal analysis techniques that combine text analysis with other modalities such as images, videos, and audio.
3. Collaboration with Online Platforms: Collaborate with social media platforms and online communities to integrate the cyberbullying detection system into their existing moderation systems.

References

1.0 *Mishra, S., Choudhary, S., Singh, S. P., & Singh, K. (2020). Detection of cyberbullying using sentiment analysis and machine learning techniques: a review. Social Network Analysis and Mining, 10(1), 1-22.*

2.0 *Kavitha, M., & Sharmila, M. (2021). Cyberbullying detection using machine learning techniques and sentiment analysis: a review. Journal of Ambient Intelligence and Humanized Computing, 1-18.*

3.0 *Rajput, R., & Sharma, S. (2020). Cyberbullying detection using machine learning and sentiment analysis. 2020 2nd International Conference on Computing, Communication, and Security (ICCCS), 104-108.*

4.0 *Arakawa, Y., & Mihara, H. (2020). Automatic cyberbullying detection system using sentiment analysis and machine learning. Multimedia Tools and Applications, 79(17-18), 11915-11933.*

5.0 *Singh, D., & Rai, N. (2021). A review on cyberbullying detection using sentiment analysis and machine learning techniques. Computer Science Review, 40, 100340.*

6.0 *Kowsari, K., Jafari Meimandi, K., Heidarysafa, M., Mendu, S., & Barnes, L. (2019). Text classification algorithms: A survey. Information, 10(4), 150. https://doi.org/10.3390/info10040150*

7.0 *Koulouri, T., Vakali, A., & Kitsios, F. (2017). Cyberbullying Detection in Social Networks Using Machine Learning Techniques. In Proceedings of the 8th Balkan Conference in Informatics (pp. 51-56). ACM. https://doi.org/10.1145/3132747.3132787*

8.0 *Mishra, N., & Mishra, A. (2018). Cyber Bullying Detection Using Machine Learning Techniques: A Survey. In Proceedings of the 3rd International Conference on Information Management and Machine Intelligence (pp. 129-138). Springer. https://doi.org/10.1007/978-981-13-0312-0_14*

9.0 *Rizvi, S. T. R., & Khan, S. (2019). A Survey of Sentiment Analysis Techniques for Social Media Posts. In Proceedings of the 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE. https://doi.org/10.1109/ICCCNT45670.2019.8944345*

10.0 *Zhang, Y., & Wang, W. (2020). A review of sentiment analysis research in social media. Journal of Ambient Intelligence and Humanized Computing, 11(4), 1511-1521. https://doi.org/10.1007/s12652-019-01412-6*

Bibliography

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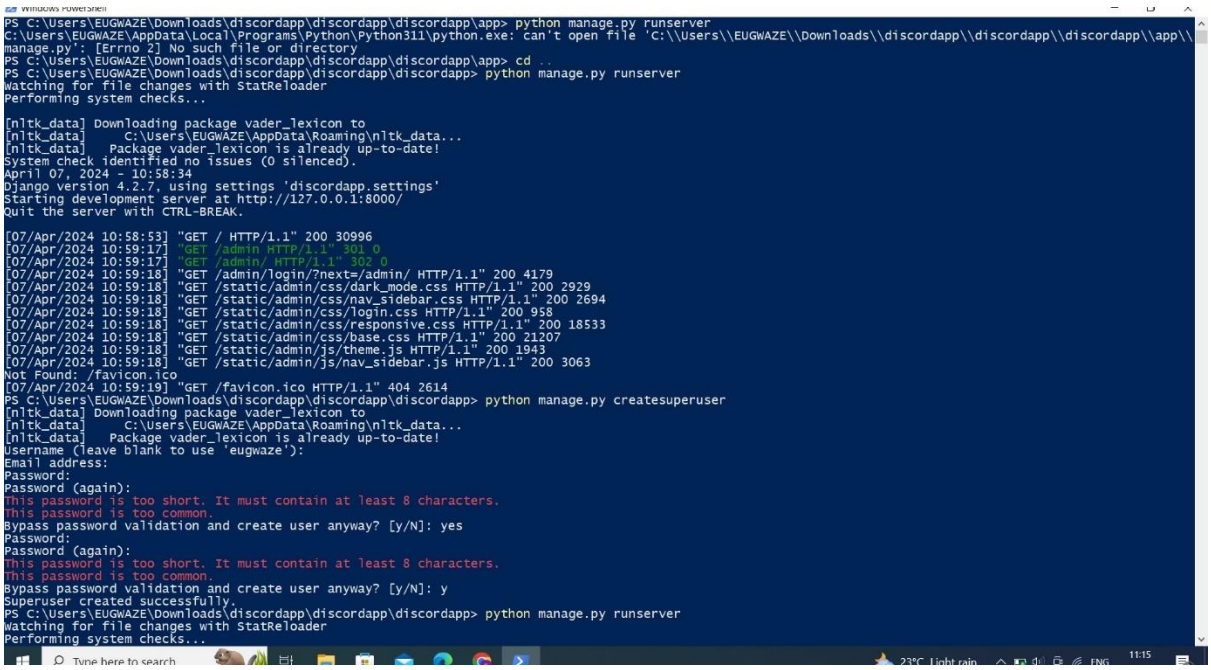
Appendix A

APPENDIX I – User Manual

Starting The System

- Open the Powershell terminal.
- Run the command "python manage.py runserver" on Powershell terminal in side the Discord folder.



```
PS C:\Users\EUGWAZE\Downloads\discordapp\discordapp\discordapp\app> python manage.py runserver
C:\Users\EUGWAZE\AppData\Local\Programs\Python\Python311\python.exe: can't open file 'C:\\Users\\EUGWAZE\\Downloads\\discordapp\\discordapp\\discordapp\\app\\
manage.py': [Errno 2] No such file or directory
PS C:\Users\EUGWAZE\Downloads\discordapp\discordapp\discordapp\app> cd ..
PS C:\Users\EUGWAZE\Downloads\discordapp\discordapp\discordapp> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\EUGWAZE\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
System check identified no issues (0 silenced).
April 07, 2024 - 10:58:34
Django version 4.2.7, using settings 'discordapp.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[07/Apr/2024 10:58:53] "GET / HTTP/1.1" 200 30996
[07/Apr/2024 10:59:17] "GET /admin HTTP/1.1" 301 0
[07/Apr/2024 10:59:17] "GET /admin/ HTTP/1.1" 302 0
[07/Apr/2024 10:59:18] "GET /admin/login/?next=/admin/ HTTP/1.1" 200 4179
[07/Apr/2024 10:59:18] "GET /static/admin/css/dark_mode.css HTTP/1.1" 200 2929
[07/Apr/2024 10:59:18] "GET /static/admin/css/nav_sidebar.css HTTP/1.1" 200 2694
[07/Apr/2024 10:59:18] "GET /static/admin/css/login.css HTTP/1.1" 200 958
[07/Apr/2024 10:59:18] "GET /static/admin/css/responsive.css HTTP/1.1" 200 18533
[07/Apr/2024 10:59:18] "GET /static/admin/css/base.css HTTP/1.1" 200 21207
[07/Apr/2024 10:59:18] "GET /static/admin/js/theme.js HTTP/1.1" 200 1943
[07/Apr/2024 10:59:18] "GET /static/admin/js/nav_sidebar.js HTTP/1.1" 200 3063
Not Found: /favicon.ico
[07/Apr/2024 10:59:19] "GET /favicon.ico HTTP/1.1" 404 2614
PS C:\Users\EUGWAZE\Downloads\discordapp\discordapp\discordapp> python manage.py createsuperuser
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\EUGWAZE\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
Username (leave blank to use 'eugwaze'):
Email address:
Password:
Password (again):
This password is too short. It must contain at least 8 characters.
This password is too common.
Bypass password validation and create user anyway? [y/N]: yes
Password:
Password (again):
This password is too short. It must contain at least 8 characters.
This password is too common.
Bypass password validation and create user anyway? [y/N]: y
Superuser created successfully.
PS C:\Users\EUGWAZE\Downloads\discordapp\discordapp\discordapp> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...
```

Fig 1 : Starting the Cyberbullying detection system

Log in the system through your browser

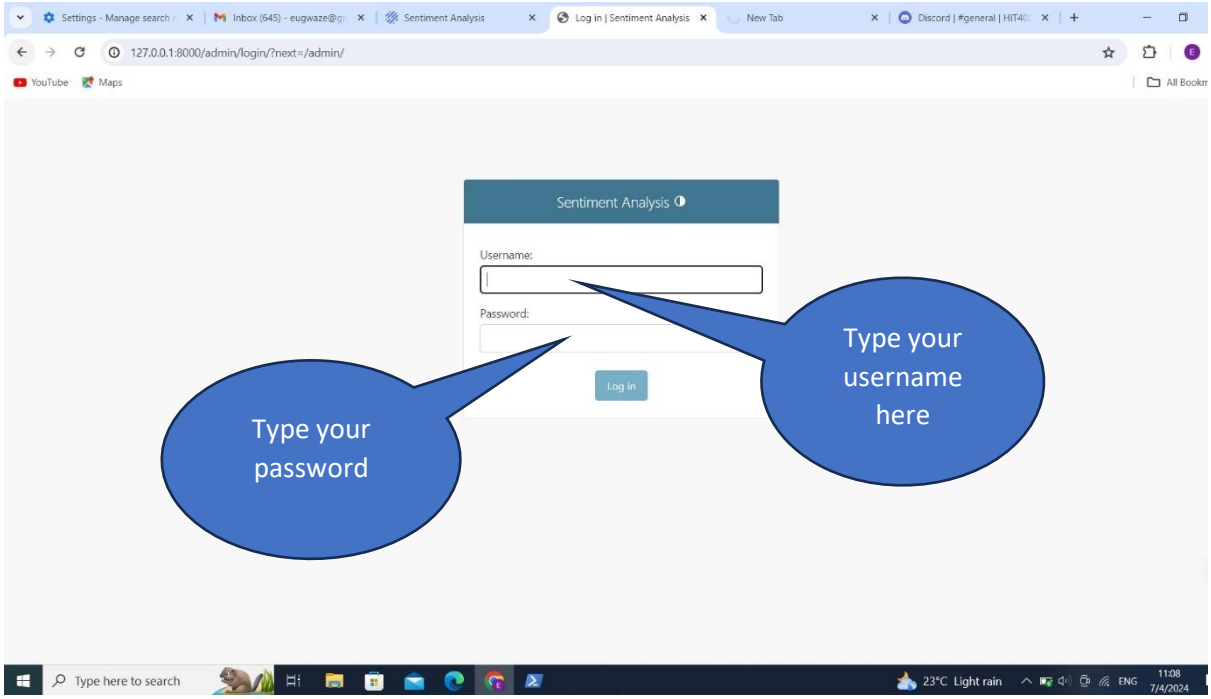


Fig 2 : Log in page

Start the Discord Server for communication

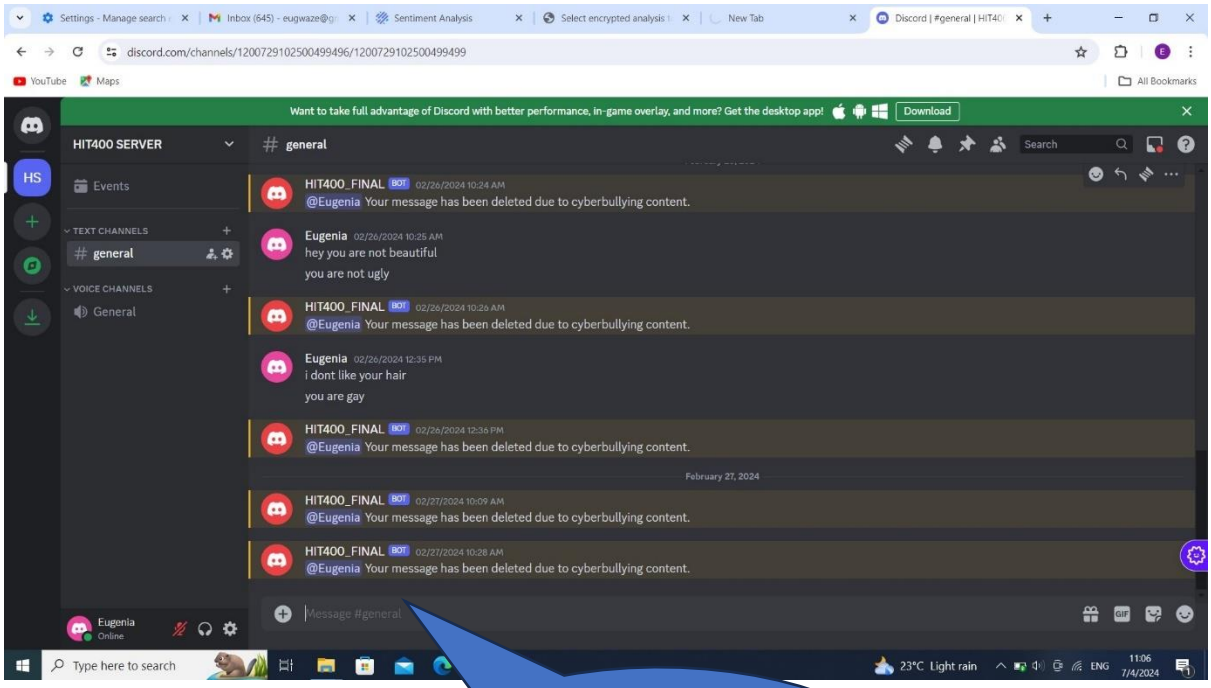


Fig 3 : Discord Server

System Database

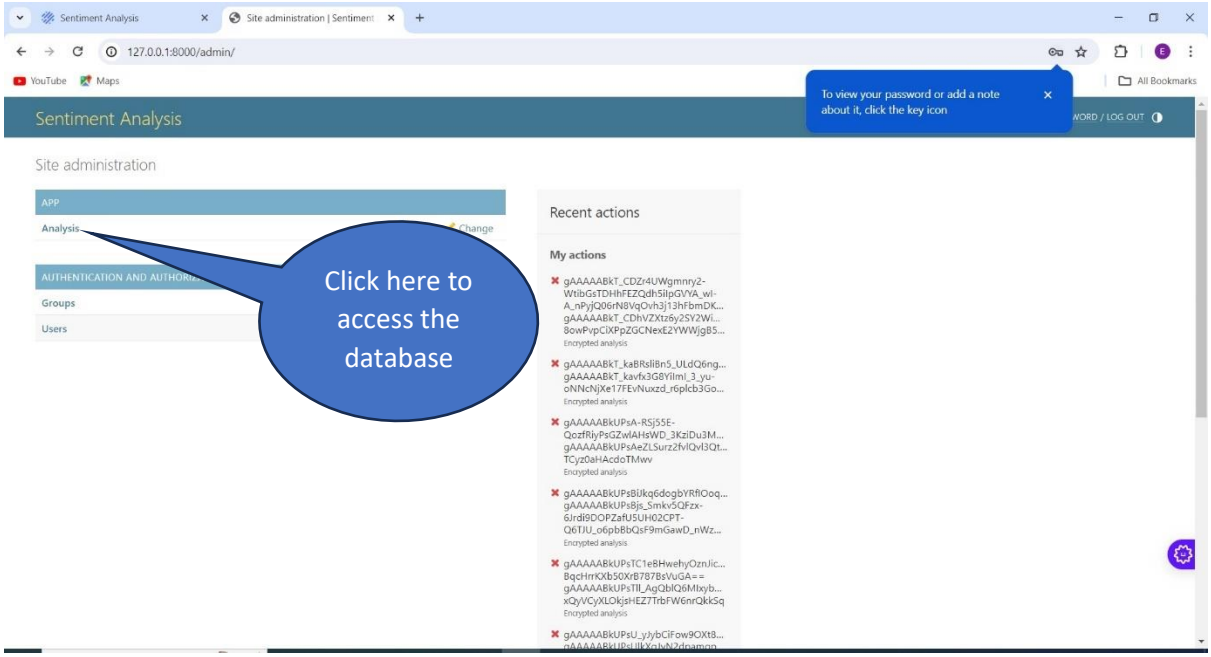


Fig 4 : Database

Aggregated Reports

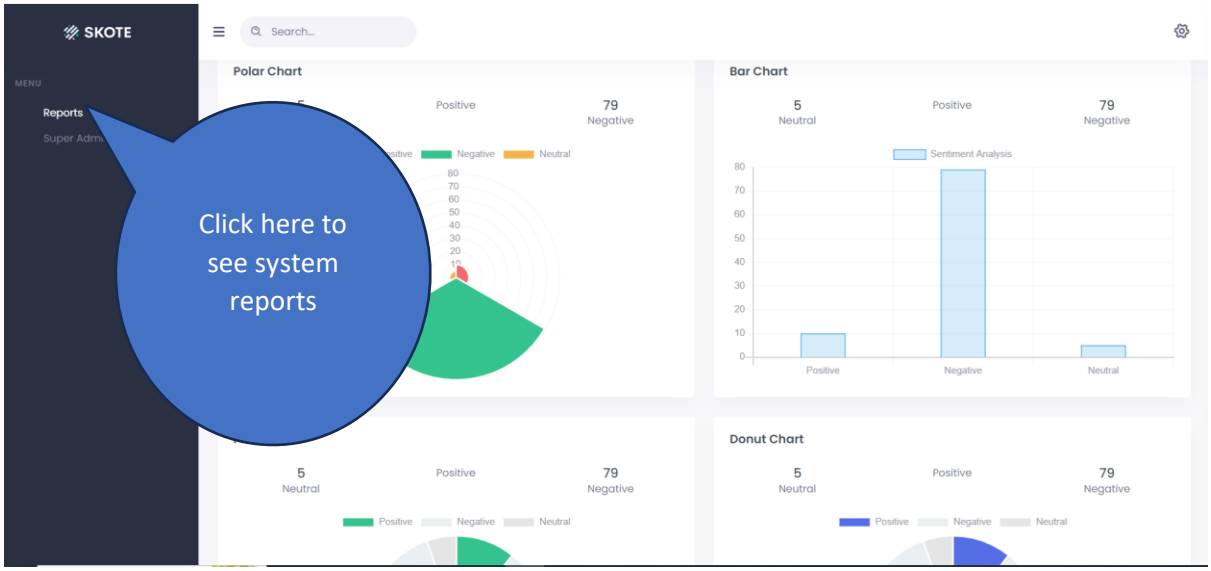


Fig 5 : Aggregated Reports

Excel Reports with sentiment score

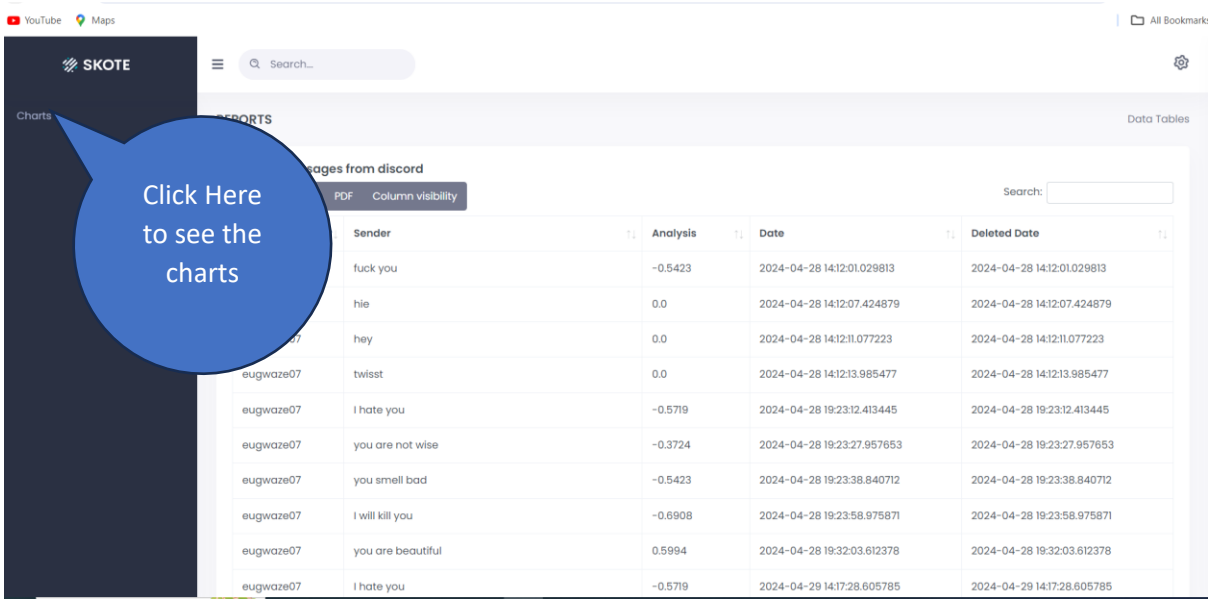


Fig 6 : Excel Report

Stopping The System

Run the command “ctrl + c” on Powershell terminal in side the Discord folder.

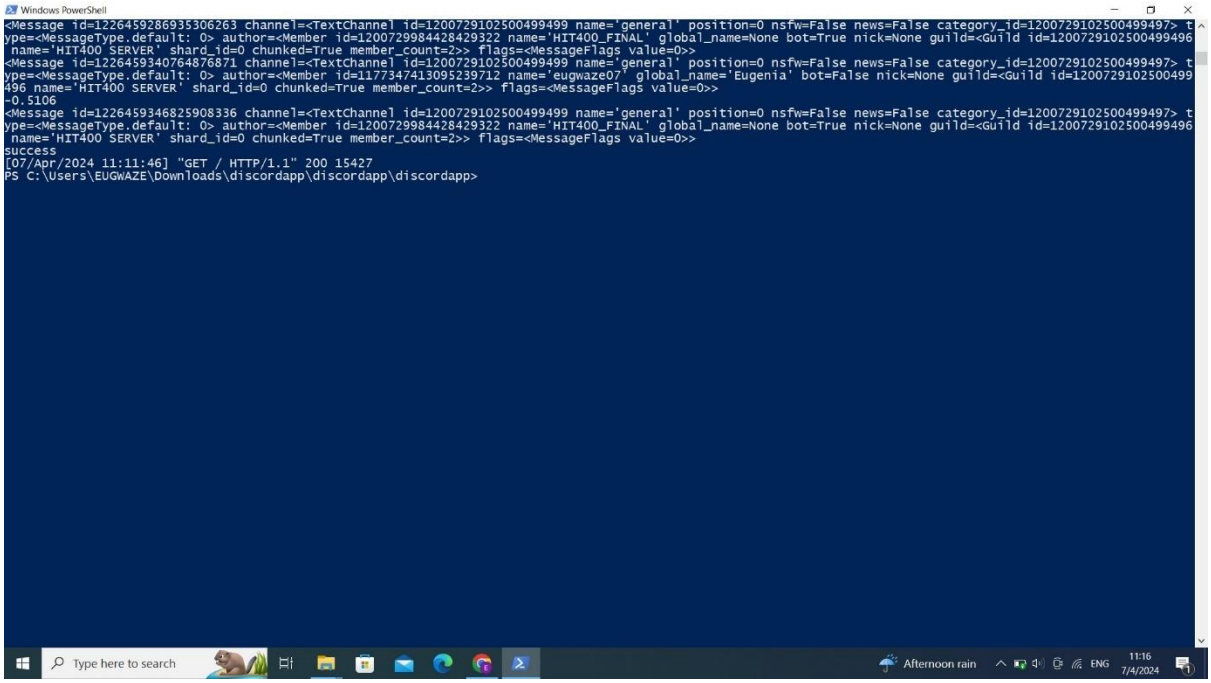


Fig 7 : Stopping the system

APPENDIX B

Appendix II : Technical Paper

CYBERBULLYING DETECTION SYSTEM

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Abstract

Cyberbullying has become a concerning issue in online game chat rooms. To tackle this problem, I have propose a Cyberbullying Detection System that uses Sentiment Analysis to identify cyberbullying behaviour in English language chat rooms. The primary objective of this system is to create a safe and secure online gaming environment by analyzing chat room messages for sentiment and producing activity reports.The system architecture has been meticulously designed to perform unsupervised and temporal sentiment analysis on text messages. This enables the early detection of cyberbullying instances and prompt implementation of preventative measures. Moreover, the system has robust data security features such as key encryption and master signing mechanisms to safeguard user data. Overall, this system has the potential to significantly improve the safety and security of online game chat rooms by providing a strong defense against cyberbullying and ensuring the well-being of users in these virtual spaces.

Keywords: sentiment analysis ,Natural language processing, Machine learning, polarity score, profanity analysis, cyberbullying

I. INTRODUCTION

The surge of cyberbullying in online game chat rooms has become a major issue, requiring the development of effective strategies for detection and prevention. This thesis proposes a Cyberbullying Detection System that uses Sentiment Analysis to identify instances of cyberbullying in English language game chat rooms. The system's main objective is to establish a secure online gaming ecosystem by scrutinizing sentiments expressed in game chat rooms. The system generates comprehensive reports to promote a culture of safety and responsible interaction among users. The system's architecture has two critical functions: unsupervised sentiment analysis and temporal analysis of text messages. These capabilities allow the system to identify and predict instances of cyberbullying, taking prompt preventative measures to mitigate harm and foster a respectful environment. To safeguard user data, the system has robust data security features such as key encryption and master signing mechanisms that protect user information from potential threats or breaches.

In summary, this visionary Cyberbullying Detection System has the potential to elevate safety and security standards within online game chat rooms, serving as a significant defense against cyberbullying. By doing so, this project takes a step towards preserving the psychological and emotional well-being of users in virtual spaces, highlighting the importance of this pioneering endeavour in the digital landscape.

II. PROBLEM STATEMENT

This project aims to create a unique Cyberbullying Detection System that can identify instances of cyberbullying in the English language. This system has the potential to significantly impact the safety and well-being of users in online game chat rooms. The main purpose of this system is to act as a vigilant guardian, constantly monitoring and responding to cyberbullying incidents. With the help

of advanced technological tools, such as sentiment analysis and real-time monitoring, it can provide a comprehensive shield against harmful online behaviours.

In the virtual world, anonymity and distance can often lead to fear and intimidation. This system aims to create a safe and secure environment for users to interact in, free from any such fears. It seeks to enrich their online gaming experience by fostering an environment where they can interact without any fear of cyberbullying.

III. RELATED WORK

There are various content filtering software systems available in the market that can be used by parents to monitor their children's social media activity and help detect and prevent cyberbullying. Some of the popular software systems include BullyBlocker, SafeChat, Facebook WatchDog, Rethink, Bark, and Bull Stop. These software systems are specifically designed to filter and analyze the online content to ensure the safety of children from online harassment and cyberbullying.

Pocket Guardian

The Pocket Guardian Application is a mobile app that was created for iOS and Android by Pocket Guardian LLC (Llc 2018). For Android devices, the Google Play Store offers the application for download, and for iOS devices, iTunes offers the same. Parents can monitor their children's social media accounts and cell phones without violating their privacy by using the Pocket Guardian program. This app will only notify parents in the event that they find evidence of sexual or cyberbullying on their children's cell phone.

SafeChat

SafeChat is a software tool that helps protect children's communication on Facebook from explicit messages. The creators of SafeChat have developed a powerful model that uses natural language processing algorithms to identify specific words and phrases that are commonly associated with explicit content.The SafeChat model works by analyzing the text of Facebook messages and searching for keywords and phrases that may indicate the presence of explicit content. If the model detects any such content, it immediately blocks the message before it can be delivered to the recipient.

to ensure that they can communicate safely and securely on Facebook.

Bully Alert

The Bully Alert app is a mobile application developed by the CU Cyber Safety Lab as a research project for the University of Colorado Boulder (Lab 2018). This app is designed for Android and can be downloaded from the Google Play Store. It allows anyone to monitor another person's social media account and receive notifications if cyberbullying is detected on that account. Currently, the app only works with Instagram accounts. To use it, simply input the Instagram profile name of the account you want to monitor into the app. The app will then begin monitoring the profile and send a notification if any

cyberbullying occurs on that account. The notification will provide details and context about the message being sent from that Instagram profile.

FacebookWatchDog

Facebook WatchDog is a software tool designed to detect threats and cyberbullying on social media. Its creators have developed a powerful system that uses social media analytics, image analysis, and text mining to identify potential threats and cyberbullying on platforms like Facebook.The tool analyzes social media activity by looking for patterns of behavior that may indicate the presence of a threat or instance of cyberbullying. This includes analyzing posts and comments for specific keywords and phrases that are commonly associated with threatening behavior or cyberbullying.

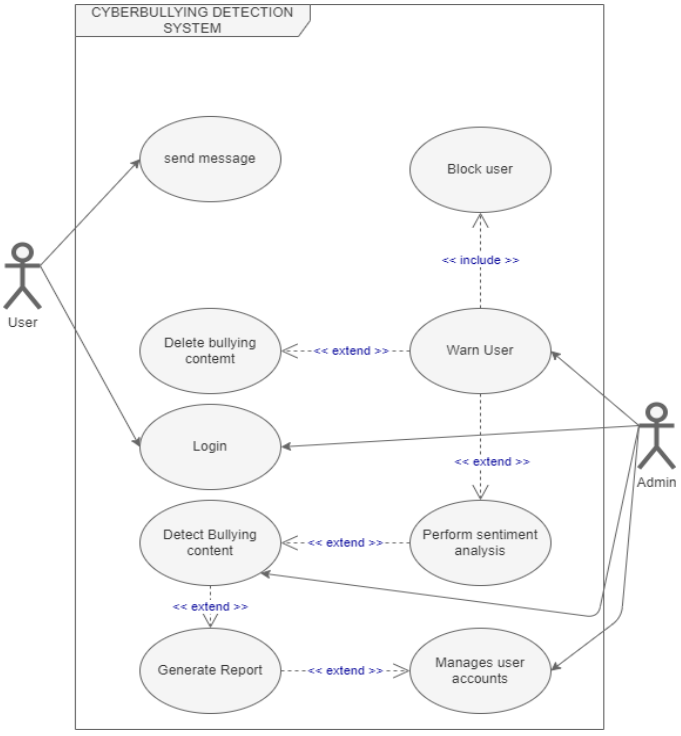


Figure 1: Use case definition of the provides a high-level overview of the interactions and functionalities of the cyberbullying detection system.

IV.SOLUTION

The proposed system is aimed at detecting cyberbullying through sentiment analysis, making use of machine learning technology. The system will consist of three main modules: the machine learning model, the chat module (based on Discord), and the admin module for reports and database management

To realize this ambitious goal, the following precise objectives have been delineated:

- To perform sentiment analysis on text messages to predict cyberbullying in game chatrooms
- To notify the Admnistrator when cyberbullying text is detected.
- To block and delete cyberbullying text
- To generate a report of the activites in the system

The proposed system would operate in the following manner:

1. The Admin creates a game room server and a chatbot that automatically listens and monitors every activity in the server. The bot API is then assigned to our system.

2. The bot forwards every message to the system, which analyzes the polarity of the message and its intent. If both are negative, the system prompts the bot to delete the messages and record the message and user of the message that was flagged and deleted into the database.

3. The system generates reports that display all the flagged messages, their users, and timeframes.

A. Features of the System

The system's core features encompass:

- Sentiment analysis within online gaming chat rooms.
- Real-time reporting and monitoring of user interactions.
- Unsupervised and temporal sentiment analysis for cyberbullying prediction.

B. Benefits of the System

The proposed system offers the following benefits:

- Comprehensive reports detailing user activities within the system.
- Real-time reporting and monitoring of user interactions.
- Unsupervised and temporal sentiment analysis for cyberbullying prediction.
- Robust data security mechanisms to safeguard user information.

C. Solution Architecture

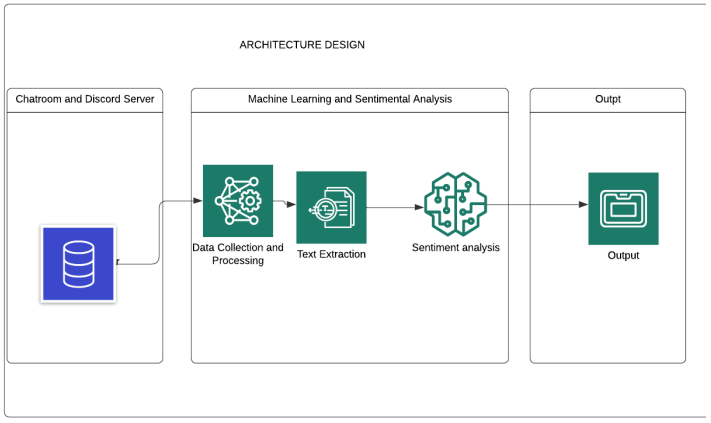
The cyberbullying detection system aims to identify and flag instances of cyberbullying in online content.

The system utilizes a combination of Naive Bayes, profanity analysis, and TextBlob analysis to achieve accurate and efficient detection. The following is a high-level solution architecture for the system:

- Data Collection: This phase is for gathering a diverse dataset consisting of online content (e.g., social media posts, comments, messages) that includes both cyberbullying and non-cyberbullying instances.
- Preprocessing: Cleaning the collected data by removing irrelevant information, such as URLs, emojis, and special characters and spliting the preprocessed dataset into training and testing sets.
- Naive Bayes Classifier: Naive Bayes algorithm to calculates the probabilities of an instance belonging to the cyberbullying or non-cyberbullying class.
- Profanity Analysis: Filters the data that can identify offensive and inappropriate language commonly associated with cyberbullying.
- TextBlob Analysis: Utilizes TextBlob, a Python library for natural language processing, to perform sentiment analysis and subjectivity analysis on the text. Sentiment analysis determines the polarity (positive, negative, or neutral) of the text.
- Aggregation and Decision Making: Combine the outputs from Naive Bayes, profanity analysis, and TextBlob analysis to arrive at a final decision.

Integration and Deployment: Integrating the cyberbullying detection system on the Discord server where there is a bot that listens to every text message that is sent in real time.

Architectural Design



This architectural design provides a high-level overview of the main components and their interactions in the cyberbullying detection system

RESULTS AND FUTURE WORKS

TABLE OF RESULTS

Domain	**Expected Results**	**Actual**
Functional Testing	• The system modules should function as expected. •The system should be easily accessible and user friendly. •Error messages should	As expected
Integration Testing	All the integrated modules should work	As expected
System Testing	All the components of the system should	As expected
Acceptance Testing	The system should meet user requirements and	As expected

Model Performance

Data Overview

- Dataset: Messages, 1,000 labelled as cyberbullying.
- Train-Test Split: 7,000 training, 3,000 testing.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AU
Naive Bayes	0.88	0.85	0.80	0.82	0.89
Profanity Analysis	0.70	0.65	0.75	0.70	0.72
TextBlob (Sentiment)	0.83	0.80	0.78	0.79	0.85
Baseline (Majority)	0.90	0.50	0.50	0.50	0.50

Error Analysis

- **False Positives**: Commonly found in sarcastic comments without explicit profanity.
- **False Negatives**: Often occur in subtle cyberbullying instances using indirect language.

Future Works

Some potential areas of future work for the system could include:

1. Expanding the system to be multilingual that is detecting cyberbullying in many languages

2. Multimodal Analysis: Incorporate multimodal analysis techniques that combine text analysis with other modalities such as images, videos, and audio.

3. Collaboration with Online Platforms: Collaborate with social media platforms and online communities to integrate the cyberbullying detection system into their existing moderation systems.

CONCLUSION

In conclusion, the "Cyberbullying Detection System" project successfully developed an automated system capable of detecting cyberbullying instances in online platforms. The system's deployment has contributed to creating a safer online environment by promptly identifying and addressing instances of cyberbullying, thus helping protect individuals from online harassment and abuse.

ACKNOWLEDGEMENT

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Lastly, I also extend my gratitude to my fellow classmates and friends who were there for me during this journey.

References

1. *Mishra, S., Choudhary, S., Singh, S. P., & Singh, K. (2020). Detection of cyberbullying using sentiment analysis and machine learning techniques: a review. Social Network Analysis and Mining, 10(1), 1-22.*
2. *Kavitha, M., & Sharmila, M. (2021). Cyberbullying detection using machine learning techniques and sentiment analysis: a review. Journal of Ambient Intelligence and Humanized Computing, 1-18.*
3. *Rajput, R., & Sharma, S. (2020). Cyberbullying detection using machine learning and sentiment analysis. 2020 2nd International Conference on Computing, Communication, and Security (ICCCS), 104-108.*
4. *Arakawa, Y., & Mihara, H. (2020). Automatic cyberbullying detection system using sentiment analysis and machine learning. Multimedia Tools and Applications, 79(17-18), 11915-11933.*
5. *Singh, D., & Rai, N. (2021). A review on cyberbullying detection using sentiment analysis and machine learning techniques. Computer Science Review, 40, 100340.*
6. *Kowsari, K., Jafari Meimandi, K., Heidarysafa, M., Mendu, S., & Barnes, L. (2019). Text classification algorithms: A*

survey. Information, 10(4), 150. https://doi.org/10.3390/info10040150

7. *Koulouri, T., Vakali, A., & Kitsios, F. (2017). Cyberbullying Detection in Social Networks Using Machine Learning Techniques. In Proceedings of the 8th Balkan Conference in Informatics (pp. 51-56). ACM. https://doi.org/10.1145/3132747.3132787*
8. *Mishra, N., & Mishra, A. (2018). Cyber Bullying Detection Using Machine Learning Techniques: A Survey. In Proceedings of the 3rd International Conference on Information Management and Machine Intelligence (pp. 129-138). Springer. https://doi.org/10.1007/978-981-13-0312-0_14*
9. *Rizvi, S. T. R., & Khan, S. (2019). A Survey of Sentiment Analysis Techniques for Social Media Posts. In Proceedings of the 2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE. https://doi.org/10.1109/ICCCNT45670.2019.8944345*
10. *Zhang, Y., & Wang, W. (2020). A review of sentiment analysis research in social media. Journal of Ambient Intelligence and Humanized Computing, 11(4), 1511-1521. https://doi.org/10.1007/s12652-019-01412-6*